SWEEP-Net: A Latent-Variable Derivation of the SEED-V Cross-Subject EEG Emotion Objective

Track A: Derivation Owner (Elnur)
Track B: Real-Analysis Certification (Nur)

September 28, 2026 (Revised Implementation & Proof Specification)

Contents

1 Introduction **4**
1.1 Two Pathologies . . . 4
1.2 Known-Dead Configurations . . . 4
1.3 The VAE Template . . . 4
1.4 Payoff . . . 5

2 Notation and Unified Geometry **5**

3 The Generative Model **5**

4 The ELBO via Jensen's Inequality **6**
4.1 The Intractable Marginal . . . 6
4.2 Introducing the Variational Posterior . . . 6
4.3 Applying Jensen's Inequality . . . 6
4.4 The Four-Term ELBO . . . 6
4.5 The Exact Nullspace E-Step . . . 7

5 The Emotion Term: OT as M-Step **7**
5.1 From Likelihood to Matching in the Closed Unit Ball $\mathbb{B}^d$ . . . 7
5.2 Normalisation and the Two Clouds . . . 8
5.3 Coupling, Dropping, and the Augmented Dustbin Matrix . . . 8
5.4 The Canonical Drop Price $\tau_y = -\varepsilon \log(1 - \rho_y)$ . . . 8
5.5 Entropy Regularisation and the Sinkhorn Mechanism . . . 9
5.6 The Matching Value and Debiasing . . . 9

6 The Subject Term: Nullspace Projection **11**
6.1 Why a Point, Not a Cloud . . . 11
6.2 From Gaussian to Weighted Least Squares . . . 11
6.3 The M-Step over the Subject Mean . . . 11
6.4 The Low-Rank Model and Global Centering . . . 11
6.5 The M-Step over U: Between-Subject Scatter and Refresh Cadence . . . 12
6.6 The Projection is the M-Step Readout (and Zero-Shot Unseen Subject Removal) . . . 12

7 The Informativeness Anchor (Pure Projected Anchor) **13**
7.1 Mutual Information of the Subject-Free Representation as the Target . . . 13
7.2 Optimal Real-vs-Random Score . . . 13
7.3 The Coupled InfoNCE Certificate . . . 13
7.4 The Decoupled (DCL) Form . . . 14

8 Gradient Analysis and Coefficient Design on the Closed Unit Ball $\mathbb{B}^d$ **15**
8.1 Per-Query Gradient Bound on $\|\tilde{z}_i\| \leq 1$ . . . 15
8.2 Two Zero-Gradient Regimes (Anti-Ghost Mechanism) . . . 15
8.3 The Gradient-Norm Ratio and λ_{OT} . . . 16
8.4 Stop-Gradient on the Coupling is Exact . . . 16

9 Queue Estimator Design **17**
9.1 The Two Sides and the Zero-Gradient Proof . . . 17
9.2 One Queue, Two Readouts . . . 17
9.3 Convergence of the β-Weighted Readout . . . 17
9.4 Leakage Guard and Staleness Bound . . . 17
9.5 The Burst Prior ρ_y: Seeding and Refinement . . . 18
9.6 Queue Size $N_q^\star$ . . . 18
9.7 The Readiness Gate . . . 19

10 Phase 1B: The Sufficient Statistic Readout and MIL Adapter **19**
10.1 The Label-Free Nullspace Marginal E-Step $\alpha_t^{(1\mathrm{B})}$ and Two-Step Baseline Contraction 19
10.2 Length-Normalised Sufficient Statistic Readout R_b . . . 20

11 The Assembled Objective **21**
11.1 The Gate (Restated) . . . 21
11.2 The Phase 1B Readout and Classifier (Restated) . . . 22
11.3 Why Memorisation is Impossible by Construction . . . 22
11.4 Practical Engineering Guards (Preventing Secondary Instabilities) . . . 22
11.5 Phase 1A Diagnostic Metrics, Remediation Ladder, and Fast Smoke-Test Protocol . 22

12 Analytic Certification (Track B) **24**
12.1 Well-Posedness of the Sparsity Term and Baseline Contraction 24
12.2 ε-Admissibility Interval 24
12.3 Definiteness of S_ε 24
12.4 Limit Behaviours 25
12.5 Non-Degeneracy (Collapse Prevention) 25
12.6 Boundedness and Lipschitz Continuity of All Terms on $\mathbb{B}^d$ 26

13 Failure Catalogue **26**

14 Conclusion **26**

Appendix: Track B Proof Obligations (1D-First Guide for Nur) **27**

Abstract

We derive, from a latent-variable generative model over EEG window embeddings, a single discriminative loss whose terms are exactly the assembled Phase-1A/1B objective of SWEEP-Net: decoupled contrastive learning (DCL) on the subject-free nullspace, Sinkhorn-infused positive matching, class-conditional unbalanced optimal transport, per-class binary-divergence sparsity, and nullspace subject projection. The derivation follows the VAE template—introduce a variational posterior, apply Jensen, obtain a bound—but the latent model serves purely as scaffolding; no reconstruction is performed. We prove that optimizing the resulting loss is, by construction, a solution to subject-ID memorization (both the anchor and the emotion prototype operate in the subject-free nullspace and carry no subject index; subject signal is confined to a low-rank subspace that is projected out), and that Phase 1B's length-normalized classification readout is the deterministic sufficient statistic of the same model evaluated via a closed-form label-free nullspace marginal E-step and a single linear Bayes plug-in head. Analytic certification of well-posedness, boundedness, definiteness, non-degeneracy, contraction of the two-step baseline estimator, and the admissible range of the entropic regulariser ε is provided in the accompanying real-analysis appendix (Track B), structured so every proof reduces to 1D real analysis and finite sums.

1 Introduction

1.1 Two Pathologies

Cross-subject EEG emotion recognition on the SEED-V 5-class benchmark is afflicted by two coupled failures:

(i) **Subject-ID memorisation ("Spatial Skull" + "Temporal Tempo").** The encoder fits skull geometry, electrode impedance, and individual alpha frequency (IAF) instead of affective content. A linear subject probe already cuts from 91.35% to 64.10% after an earlier alignment attempt, yet cross-subject accuracy plateaus at 29–32%.

(ii) **Gaslighting (temporal label pollution).** A 60 s Fear/Disgust trial contains ∼5–10 s of genuine affective burst buried in ∼50 s of baseline. Window-level global labels entangle baseline with burst.

1.2 Known-Dead Configurations

The following approaches have been tried and are provably or empirically dead; the derivation must not reduce to any of them:

- **cMMD** (class-conditional MMD, two movable distributions, no anchor) → constant-vector collapse.
- **DANN / GRL** (min–max saddle) → NaN blowup (7×10^{15} gradient norm).
- **Balanced OT for alignment** → SCDA subject-mining pathology (87% subject probe).
- **Back-prop through the Sinkhorn fixed point; queue-vs-queue OT** (zero-gradient path).

1.3 The VAE Template

The VAE derivation proceeds: want $\log p(x)$; face an intractable marginal; introduce a variational posterior; apply Jensen; obtain the ELBO; optimise it. We do the same. *We are not generating or reconstructing EEG.* The latent model is scaffolding to derive the discriminative loss.

1.4 Payoff

The final loss is our current objective, and the derivation proves:

(a) Both the emotion term and the Phase 1B gate depend only on nullspace-projected, subject-free coordinates with no subject index $\implies$ memorisation is unrewarded.

(b) Subject signal is confined to a low-rank term that is projected out without requiring test-time subject adaptation.

(c) Phase 1B's single sufficient-statistic readout is the deterministic readout of the same model.

2 Notation and Unified Geometry

Symbol	Meaning
b	A bag (trial); label $y_b \in \{1, \dots, C\}$ with $C = 5$; subject $s \in \{1, \dots, S\}$
x_t	1-second window, 32 time-steps, 5-band CWT topo-map
$z_t = f(x_t) \in \mathbb{S}^{d-1}$	L2-normalised encoder embedding in unified dimension d ($d = 256$)
$h_t = z_t - \mu \in \mathbb{R}^d$	Globally centred window embedding (μ is the global baseline mean)
$h'_t = (I - UU^\top)h_t \in \mathbb{R}^d$	Subject-nullspace projected embedding
$\tilde{z}_t = \frac{h'_t}{\max(\|h'_t\|, \tau_{\text{norm}})} \in \mathbb{B}^d$	Norm-floored nullspace feature in the closed unit ball $\mathbb{B}^d = \{v \in \mathbb{R}^d : \|v\| \le 1\}$ ($\tau_{\text{norm}} = 0.25$)
$e_t \in \{0, 1\}$	Latent burst indicator for window t
$\rho_y \in [\rho_{\min}, 1 - \rho_{\min}]$	Per-class burst probability (small for Fear/Disgust; $\rho_{\min} > 0$)
p_y	Emotion prototype cloud in $\text{span}(U)^\perp \cap \mathbb{B}^d$ (**no subject index**)
$\mu_s = \mu + Ua_s$	Subject baseline mean (**no emotion index**)
$\alpha_t = q(e_t = 1 \mid x_t)$	Exact posterior responsibility (E-step), $\alpha_t \in [\alpha_{\min}, \alpha_{\max}] \subset (0, 1)$
T_b	Number of 1-second windows in bag b

Remark 2.1 (Single Unified Embedding Space and Closed Unit Ball Domain $\mathbb{B}^d$). To prevent representation splitting between Phase 1A and Phase 1B ("two geometries, one gate"), all objects—the momentum queue keys $r_j \in \mathbb{B}^d$, the projected DCL anchor $\mathcal{L}_{\text{anc}}(\tilde{z}_t)$, the OT cost C_{ij}, the between-subject scatter $S_{\text{between}} \in \mathbb{R}^{d \times d}$, the projector $I - UU^\top \in \mathbb{R}^{d \times d}$, and the Phase 1B readout $R_b \in \mathbb{R}^d$—live in the **same** d-dimensional space ($d = 256$). Because the safe norm floor $\tilde{z}_t = h'_t / \max(\|h'_t\|, \tau_{\text{norm}})$ satisfies $\|\tilde{z}_t\| = 1$ in the active regime $\|h'_t\| \ge \tau_{\text{norm}}$ and $\|\tilde{z}_t\| = \|h'_t\|/\tau_{\text{norm}} < 1$ when $\|h'_t\| < \tau_{\text{norm}}$, $\tilde{z}_t$ and r_j live in the convex, compact **closed unit ball** $\mathbb{B}^d = \{v \in \mathbb{R}^d : \|v\| \le 1\}$.

3 The Generative Model

For bag b (label y, subject s), the latent-variable model is:

$$e_t \sim \text{Bern}(\rho_y), \tag{1}$$

$$z_t \mid (e_t = 1) \sim p_y \quad (\text{emotion prototype in } \text{span}(U)^\perp \text{ — no } s), \tag{2}$$

$$z_t \mid (e_t = 0) \sim \mathcal{N}(\mu_s, \Sigma_{\text{base}}) \quad (\text{subject baseline — no } y), \tag{3}$$

$$x_t \sim p_{\text{dec}}(x \mid z_t). \tag{4}$$

Remark 3.1 (Structural choices that do the heavy lifting). p_y carries no subject index and is supported on the subject-free nullspace $\text{span}(U)^\perp$; the baseline carries no emotion index. Weak labelling is built in: y is per-bag, e_t is per-window and latent.

Objective. Maximise $\sum_b \log p(x_b, y_b)$, marginalising over e and subject parameters.

4 The ELBO via Jensen's Inequality

4.1 The Intractable Marginal

For bag b with T_b windows, the marginal likelihood requires summing over all 2^{T_b} configurations of the latent burst indicators:

$$p(x_b, y_b) = \sum_{e_1=0}^{1} \cdots \sum_{e_{T_b}=0}^{1} \prod_{t=1}^{T_b} p(x_t \mid z_t)\, p(z_t \mid e_t, y_b, s)\, p(e_t \mid y_b).$$

This is intractable.

4.2 Introducing the Variational Posterior

Introduce the factorised variational posterior

$$q(e \mid x) = \prod_{t=1}^{T_b} q(e_t \mid x_t) = \prod_{t=1}^{T_b} \mathrm{Bern}(\alpha_t),$$

where $\alpha_t = q(e_t = 1 \mid x_t)$.

4.3 Applying Jensen's Inequality

Theorem 4.1 (Weighted Jensen Bound)**.** *Let $\{A_e\}_{e \in E}$ be positive reals and $q = (q_e)$ a weight vector ($q_e \geq 0$, $\sum_e q_e = 1$). Then*

$$\log\left(\sum_e A_e\right) \geq \sum_e q_e \log \frac{A_e}{q_e} = \sum_e q_e \log A_e + H(q),$$

where $H(q) = -\sum_e q_e \log q_e$ is the entropy, with equality if and only if $q_e = A_e / \sum_{e'} A_{e'}$.

[Track B Proof Guide (for Nur) — B1 — Weighted Jensen Bound via 1D Calculus. Step 1 (1D inequality): Define the single-variable function $g(t) = \log t - (t-1)$ for $t > 0$. Compute $g'(t) = \frac{1}{t} - 1$ and $g''(t) = -\frac{1}{t^2} < 0$. Show $t = 1$ is the unique global maximum on $(0, \infty)$, so $\log t \leq t - 1$ with equality iff $t = 1$.
Step 2 (2-term then n-term sum): Let $S = \sum_e A_e$ and $p_e = A_e/S$ (so $\sum_e p_e = 1$). Substitute $t = p_e/q_e$ into $\log t \leq t - 1$, multiply by $q_e > 0$, and sum over e:

$$\sum_e q_e \log \frac{p_e}{q_e} \leq \sum_e q_e \left(\frac{p_e}{q_e} - 1\right) = \sum_e p_e - \sum_e q_e = 1 - 1 = 0.$$

Substitute $p_e = A_e/S$ and use $\log(A_e/(Sq_e)) = \log(A_e/q_e) - \log S$ to rearrange into Theorem 4.1. Equality holds iff $p_e/q_e = 1$ for every e, i.e., $q_e = A_e/S$.]

4.4 The Four-Term ELBO

Applying Theorem 4.1 to the marginal likelihood, the ELBO decomposes into four terms:

$$\log p(x, y) \geq \underbrace{\sum_t \alpha_t \log p_y(\tilde{z}_t)}_{\text{emotion}} + \underbrace{\sum_t (1 - \alpha_t) \log p_s(z_t)}_{\text{subject}} - \underbrace{\sum_t \mathrm{KL}\big(q(e_t) \parallel \mathrm{Bern}(\rho_y)\big)}_{\text{sparsity}} + \underbrace{I(\tilde{z}; x)}_{\text{projected anchor}}. \tag{5}$$

Derivation. By Jensen's inequality applied to the concave function log:

$$\log p(x,y) = \log \sum_e p(x,y,e) = \log \sum_e q(e) \frac{p(x,y,e)}{q(e)}$$
$$\geq \sum_e q(e) \log \frac{p(x,y,e)}{q(e)} = \sum_e q(e) \log p(x,y,e) - \sum_e q(e) \log q(e).$$

Under the factorised model, the joint decomposes per-window. The first sum gives the emotion and subject terms (weighted by α_t and $1-\alpha_t$ respectively). The second sum gives the sparsity KL (comparing $q(e_t) = \text{Bern}(\alpha_t)$ to the prior $\text{Bern}(\rho_y)$). The projected anchor $I(\tilde{z}; x)$ ensures the nullspace-projected representation retains mutual information with x. □

4.5 The Exact Nullspace E-Step

During Phase 1A training, the bag label y is known. The posterior $\alpha_t = q(e_t = 1 \mid x_t, y)$ is available in closed form because the conditional marginal is the two-component mixture. By Bayes' rule:

$$\alpha_t^{(1A)} = \frac{\rho_y \, p_y(\tilde{z}_t)}{\rho_y \, p_y(\tilde{z}_t) + (1-\rho_y) \, p_s(h'_t)} = \sigma\big(\log p_y(\tilde{z}_t) - \log p_s(h'_t) + \text{logit}(\rho_y)\big) \,. \tag{6}$$

Proof. By Bayes' theorem for the two-component mixture:

$$\alpha_t^{(1A)} = P(e_t = 1 \mid h'_t, y) = \frac{p_y(\tilde{z}_t)\, \rho_y}{\rho_y p_y(\tilde{z}_t) + (1-\rho_y) p_s(h'_t)}.$$

Dividing numerator and denominator by $(1-\rho_y) p_s(h'_t)$ yields

$$\alpha_t^{(1A)} = \frac{\frac{\rho_y}{1-\rho_y} \cdot \frac{p_y(\tilde{z}_t)}{p_s(h'_t)}}{1 + \frac{\rho_y}{1-\rho_y} \cdot \frac{p_y(\tilde{z}_t)}{p_s(h'_t)}} = \sigma\left(\log \frac{p_y(\tilde{z}_t)}{p_s(h'_t)} + \log \frac{\rho_y}{1-\rho_y} \right). \qquad \square$$

This is the exact E-step paired with the M-step: α is a deterministic function of the current model, calibrated by construction, and never jointly gradient-descended as an independent quantity. This is the classical EM structure [1], justified because the mixture posterior is tractable [2]. For early stability, apply deterministic annealing [3]: compute α_t with the log-ratio divided by a temperature T annealed from high to low.

Remark 4.2 (Nullspace baseline score and strict interiority $\alpha_t \in [\alpha_{\min}, \alpha_{\max}]$). Because $(I - UU^\top)$ annihilates the in-span subject offset Ua_s, evaluating the baseline Gaussian density in nullspace coordinates $\log p_s(h'_t) = -\|h'_t - \hat{\mu}'_s\|^2/(2\sigma^2)$ (where $\hat{\mu}'_s = (I - UU^\top)(\mu^\star_s - \mu)$ is the out-of-subspace baseline residual) prevents within-trial electrode drift along $\text{span}(U)$ from falsely firing $\alpha_t \to 1$. Meanwhile, $\log p_y(\tilde{z}_t)$ is the compatibility score over the prototype cloud in $\mathbb{B}^d$, $\log \sum_{j \in Q_y} b_j \exp(-(1 - \tilde{z}_t^\top r_j)/\varepsilon)$. Since $\|\tilde{z}_t\| \leq 1$, $\|h'_t\| \leq 2$, and $\|r_j\| \leq 1$, both $\log p_y(\tilde{z}_t)$ and $\log p_s(h'_t)$ are uniformly bounded on $[-M, M]$. With $\rho_y \in [\rho_{\min}, 1 - \rho_{\min}] \subset (0,1)$, the total logit in (6) is bounded in $[-M_{\text{logit}}, M_{\text{logit}}]$, which guarantees

$$\alpha_t \in [\alpha_{\min}, \alpha_{\max}] \subset (0,1), \qquad \alpha_{\min} := \sigma(-M_{\text{logit}}) > 0, \quad \alpha_{\max} := \sigma(M_{\text{logit}}) < 1.$$

Consequently, neither α_t nor $1 - \alpha_t$ can ever vanish on a non-empty batch.

5 The Emotion Term: OT as M-Step

5.1 From Likelihood to Matching in the Closed Unit Ball $\mathbb{B}^d$

The emotion term of the ELBO is $\sum_i \alpha_i \log p_y(\tilde{z}_i)$. Both the current window features and the queue prototypes live in the subject-free nullspace within the closed unit ball $\mathbb{B}^d$:

$$h'_i = (I - UU^\top)(z_i - \mu), \qquad \tilde{z}_i = \frac{h'_i}{\max(\|h'_i\|, \tau_{\text{norm}})} \in \mathbb{B}^d, \qquad \tau_{\text{norm}} = 0.25.$$

With symmetric cost $C_{ij} = 1 - \tilde{z}_i^\top r_j \in [0, 2]$ on $\mathbb{B}^d \times \mathbb{B}^d$, maximising the weighted likelihood becomes minimising a transport cost.

5.2 Normalisation and the Two Clouds

For class y with index set $I_y = \{i : y_i = y\}$, define the expected burst mass:

$$A_y = \sum_{i \in I_y} \alpha_i = \mathbb{E}_q \left[\sum_{i \in I_y} e_i \;\middle|\; x \right].$$

Hence $A_y/|I_y|$ is the method-of-moments estimator of the prior burst probability ρ_y, and the sparsity term $\mathrm{KL}(q(e_i \mid x_i) \,\|\, \mathrm{Bern}(\rho_y))$ is the penalty that keeps this estimator tied to the prior.

By Remark 4.2, whenever class y is present in the batch ($I_y \neq \emptyset$), $A_y \geq |I_y|\alpha_{\min} > 0$ strictly (if $I_y = \emptyset$, the class-y emotion loss is 0). Set $a_i = \alpha_i / A_y$ and define the *current cloud*

$$\mu_y = \sum_{i \in I_y} a_i \, \delta_{\tilde{z}_i}.$$

The absolute amount stays in A_y and is regulated by the sparsity prior, while μ_y carries only the shape of the posterior burst cloud.

The *reference cloud* is the class-y slice of the queue Q_y with stop-gradient nullspace embeddings $r_j \in \mathbb{B}^d$ and normalised burst weights $b_j = \beta_j / \sum_{k \in Q_y} \beta_k \geq 0$ summing to one:

$$\nu_y = \sum_{j \in Q_y} b_j \, \delta_{r_j}.$$

5.3 Coupling, Dropping, and the Augmented Dustbin Matrix

A coupling $\Pi_{ij} \geq 0$ moves normalised mass from $\tilde{z}_i$ ($i \in \{1, \ldots, n\}$, $n = |I_y|$) to r_j ($j \in \{1, \ldots, m\}$, $m = |Q_y|$). Balanced marginals would force every bit of posterior burst mass onto the queue and every prototype to be fully used, which is wrong because the queue may be stale and some posterior burst mass may be unexplainable. We therefore relax both sides with non-negative slacks $d_i = \Pi_{i0} \geq 0$ (current mass *dropped*) and $u_j = \Pi_{0j} \geq 0$ (queue mass left *unused*):

$$\sum_{j=1}^{m} \Pi_{ij} + d_i = a_i \quad (i = 1, \ldots, n), \qquad \sum_{i=1}^{n} \Pi_{ij} + u_j = b_j \quad (j = 1, \ldots, m). \tag{7}$$

Introducing a dummy baseline node 0 on both sides with marginal capacity $a_0 = b_0 = 1$ and corner slack $\Pi_{00} = 1 - \sum_{i,j} \Pi_{ij} - \sum_i d_i - \sum_j u_j = \sum_{i,j} \Pi_{ij} \geq 0$, the constraints (7) become an **exact balanced transport problem** on the $(n+1) \times (m+1)$ augmented coupling $\tilde{\Pi} = [\Pi_{ij}]_{0 \leq i \leq n, 0 \leq j \leq m}$ with row sum $(a_0, a_1, \ldots, a_n)$, column sum $(b_0, b_1, \ldots, b_m)$, and augmented cost matrix $\tilde{C}_{ij} = C_{ij}$ for $i, j \geq 1$, $\tilde{C}_{i0} = \tilde{C}_{0j} = \tau_y$, and $\tilde{C}_{00} = 0$.

5.4 The Canonical Drop Price $\tau_y = -\varepsilon \log(1 - \rho_y)$

Dropping must be priced. In entropy-regularised OT the ratio C_{ij}/ε is dimensionless, so ε carries the cost scale and the prior burst probability $\rho_y \in (0, 1)$ enters through a dimensionless increasing function $h(\rho_y)$:

$$\tau_y = \varepsilon \, h(\rho_y), \qquad h(\rho_y) := -\log(1 - \rho_y). \tag{8}$$

Why this exact functional form is canonical:

(i) **Zero free hyperparameters and proper units:** $h(\rho_y) = -\log(1-\rho_y)$ is dimensionless and strictly increasing on $(0,1)$ since $h'(\rho_y) = \frac{1}{1-\rho_y} > 0$.

(ii) **Exact Bayesian dustbin weight:** Under the Sinkhorn Gibbs kernel $\exp(-\tilde{C}_{ij}/\varepsilon)$, the drop transition weight is

$$\exp(-\tau_y/\varepsilon) = \exp\big(\log(1-\rho_y)\big) = 1-\rho_y,$$

which is literally the prior baseline probability $P(e_t = 0 \mid y) = 1-\rho_y$.

(iii) **Exact boundary limits:** As $\rho_y \to 0^+$ (all baseline expected), $h(0^+) = 0$ and $\tau_y \to 0^+$, so dropping is free ($\exp(-\tau_y/\varepsilon) \to 1$). As $\rho_y \to 1^-$ (all burst expected), $h(1^-) = +\infty$ and $\tau_y \to +\infty$, so $\exp(-\tau_y/\varepsilon) \to 0$, forcing $d_i = u_j = 0$ and recovering **exact balanced Sinkhorn OT**.

Remark 5.1 (d_i is not the baseline route). The baseline mass $1-\alpha_i$ never enters μ_y, so the fraction of class-y windows left unmatched by construction is already $1 - A_y/|I_y|$, and the total unmatched fraction is

$$U_y = 1 - \frac{A_y}{|I_y|}(1 - D_y^{\text{norm}}), \qquad D_y^{\text{norm}} = \sum_{i \in I_y} d_i.$$

When the sparsity prior gives $A_y/|I_y| \approx \rho_y$ and the queue matches well so $D_y^{\text{norm}} \approx 0$, we recover $U_y \approx 1-\rho_y$.

5.5 Entropy Regularisation and the Sinkhorn Mechanism

To obtain smooth, strictly positive soft assignments over both matches and drops, apply the entropy regulariser to the full $(n+1)\times(m+1)$ augmented coupling $\tilde{\Pi}$ (with $0\log 0 := 0$):

$$\begin{aligned} R_\varepsilon(\tilde{\Pi}) &= \varepsilon \sum_{i=0}^{n}\sum_{j=0}^{m} \tilde{\Pi}_{ij}(\log \tilde{\Pi}_{ij} - 1) \\ &= \varepsilon \sum_{i,j\geq 1} \Pi_{ij}(\log \Pi_{ij} - 1) + \varepsilon \sum_{i=1}^{n} d_i(\log d_i - 1) + \varepsilon \sum_{j=1}^{m} u_j(\log u_j - 1) + \varepsilon \Pi_{00}(\log \Pi_{00} - 1). \end{aligned} \tag{9}$$

The minimiser carries the Gibbs factors $\exp(-C_{ij}/\varepsilon)$ for real matches and $\exp(-\tau_y/\varepsilon) = 1-\rho_y$ for dropping, solved by standard balanced Sinkhorn matrix scaling on the $(n+1)\times(m+1)$ kernel $K = \exp(-\tilde{C}/\varepsilon)$ [4, 5].

5.6 The Matching Value and Debiasing

Definition 5.2 (Regularised Unbalanced Matching Value)**.**

$$W_\varepsilon^y(\mu_y, \nu_y) := \min_{\tilde{\Pi}\geq 0} \left[\sum_{i=1}^{n}\sum_{j=1}^{m} \Pi_{ij} C_{ij} + \tau_y \sum_{i=1}^{n} d_i + \tau_y \sum_{j=1}^{m} u_j + R_\varepsilon(\tilde{\Pi}) \right]$$

subject to the slack constraints (7).

Proposition 5.3 (Symmetry of W_ε)**.** *Swapping μ_y and ν_y exchanges $d \leftrightarrow u$ and leaves the problem unchanged, so $W_\varepsilon^y(\mu_y, \nu_y) = W_\varepsilon^y(\nu_y, \mu_y)$.*

Proof. Under the swap, the constraint $\sum_j \Pi_{ij} + d_i = a_i$ becomes $\sum_i \Pi_{ji} + u_j = b_j$ (with indices relabelled), and vice versa. The cost matrix transposes: $C_{ij} \to C_{ji}$. The objective is invariant under this relabelling. □

W_ε contains self-energy and entropy blur, so a cloud matched to itself does not score zero. The debiased divergence removes this:

Definition 5.4 (Debiased Sinkhorn Divergence)**.**

$$S_\varepsilon^y(\mu_y, \nu_y) := W_\varepsilon^y(\mu_y, \nu_y) - \tfrac{1}{2}W_\varepsilon^y(\mu_y, \mu_y) - \tfrac{1}{2}W_\varepsilon^y(\nu_y, \nu_y).$$

This satisfies $S_\varepsilon^y(\mu_y, \mu_y) = 0$ and is symmetric.

Proposition 5.5 (Emotion M-Step)**.** *The emotion M-step for class y is*

$$\mathcal{L}_{\text{em}}^{(y)} = S_\varepsilon^y(\mu_y, \nu_y),$$

and the total emotion contribution is $\mathcal{L}_{\text{em}} = \sum_y S_\varepsilon^y(\mu_y, \nu_y)$, scaled by λ_{OT}. Since p_y carries no subject index and resides in $\text{span}(U)^\perp$, reducing this loss requires cross-subject burst alignment.

[Track B Proof Guide (for Nur) — B3 — Definiteness of S_ε via a 1D Two-Point Model. Step 1 (Symmetry and self-zero): Plug $a = b$ directly into Definition 5.4 to get $S_\varepsilon(a, a) = W_\varepsilon(a, a) - \frac{1}{2}W_\varepsilon(a, a) - \frac{1}{2}W_\varepsilon(a, a) = 0$, and use Proposition 5.3 for $S_\varepsilon(a, b) = S_\varepsilon(b, a)$. Step 2 (1D two-point model for non-negativity and drop saturation): Let $a = \delta_x$ and $b = \delta_y$ be two single points separated by cost $c = C(x, y) \in [0, 2]$ with drop price $\tau_y > 0$. Write the 1D function of a single variable $p = \Pi_{11} \in [0, 1]$ (the matched mass, so dropped mass on each side is $d_1 = 1 - p$ and $u_1 = 1 - p$):

$$\psi_c(p) = c\,p + 2\tau_y(1 - p) + \varepsilon\big[p(\log p - 1) + 2(1 - p)(\log(1 - p) - 1) + p(\log p - 1)\big].$$

In the $\varepsilon \to 0^+$ limit, $\min_{p\in[0,1]}[c\,p + 2\tau_y(1 - p)] = \min(c, 2\tau_y)$, while at $c = 0$ (self-matching) it is 0. Thus $S_0(\delta_x, \delta_y) = \min(c, 2\tau_y) \geq 0$: it is strictly positive for $0 < c < 2\tau_y$ (active matching regime) and saturates flat at $2\tau_y$ for $c \geq 2\tau_y$ (all mass dropped: $p^\star = 0$). For $\varepsilon > 0$, set $\psi_c'(p^\star) = 0$ using 1D calculus to show $W_\varepsilon(\delta_x, \delta_y)$ is strictly increasing in c, so $S_\varepsilon(\delta_x, \delta_y) = W_\varepsilon(c) - W_\varepsilon(0) \geq 0$.
Step 3 (Multi-point extension): State how the general cloud case sums this exact 2-point comparison over pairs (i, j), remaining positive in the active cone $C_{ij} < 2\tau_y$ and saturating at $2\tau_y$ when $\min_{i,j} C_{ij} \gg 2\tau_y$.]

[Track B Proof Guide (for Nur) — B2 — ε-Admissibility Interval via 1D Contraction and Entropy Bounds.
B2a (1D log-ratio contraction → lower bound on ε): First consider a 2×2 positive table with costs bounded by $\tilde{M} = \max(2, \tau_y)$, so entries lie in $[e^{-\tilde{M}/\varepsilon}, 1]$. Define the 1D log-ratio scaling map $f(u) = \log \frac{e^{u-\Delta}+1}{e^u+e^{-\Delta}}$ for $u \in \mathbb{R}$, where $\Delta = \tilde{M}/(2\varepsilon) > 0$. Differentiate with respect to u using single-variable calculus to show $0 < f'(u) \leq \tanh(\Delta) = \tanh(\tilde{M}/2\varepsilon) < 1$ (attained at $u = 0$). By the 1D Mean Value Theorem, each iteration shrinks the error by $\tanh(\tilde{M}/2\varepsilon)$. Requiring $[\tanh(\tilde{M}/2\varepsilon)]^T \leq \text{tol}$ and taking artanh gives $\varepsilon \geq \frac{\tilde{M}}{2\,\text{artanh}(\text{tol}^{1/T})}$. Note that the $(n + 1) \times (m + 1)$ Sinkhorn update applies this same weighted-average contraction across rows and columns.
B2b (Naive upper bound): Apply Theorem 4.1 (B1) with uniform weights $q_{ij} = \frac{1}{(n+1)(m+1)}$ to show that the negative entropy of $\tilde{\Pi}$ is bounded by $\varepsilon \log((n + 1)(m + 1))$, giving $|W_\varepsilon - W_0| \leq \varepsilon \log((n + 1)(m + 1))$.
B2c (Refined ℓ_1 blur bound): Start in 1D with two 2-bin weight vectors $a = (a_1, 1 - a_1)$ and $b = (b_1, 1 - b_1)$. Shifting mass $|a_1 - b_1|$ between bins changes the transport cost by at most $\tilde{M}|a_1 - b_1|$. Summing over bins by the triangle inequality gives $|S_\varepsilon(a, b) - W_0(a, b)| \leq (L + L_0)\sum_i |a_i - b_i| = (L + L_0)d_1(a, b)$, independent of n, m.]

6 The Subject Term: Nullspace Projection

6.1 Why a Point, Not a Cloud

The emotion branch uses a free empirical cloud matched by OT. The subject branch must not, for three reasons:

(i) **Role.** The emotion cloud p_y is shared across subjects and is something to align to. The subject baseline is a per-subject nuisance to capture and remove, not a prototype to match.

(ii) **Degeneracy.** Maximising $\sum_t (1-\alpha_t) \log p_s(z_t)$ over a free measure p_s is ill-posed: the optimum piles all mass on the single point with the largest weight.

(iii) **Sufficiency and pathology.** The baseline is tonic and roughly unimodal, so its first moment is all the subject signal that matters. Keeping a rich subject cloud would reward exactly the SCDA subject-mining pathology on the dead list.

6.2 From Gaussian to Weighted Least Squares

Take $\Sigma_{\text{base}} = \sigma^2 I$. Then, up to constants,

$$\log p_s(z_t) \propto -\frac{1}{2\sigma^2} \|z_t - \mu_s\|^2,$$

and the negative log-likelihood subject term becomes the weighted least-squares problem:

$$\mathcal{L}_{\text{sub}} = \sum_{t \in s} (1 - \alpha_t) \|z_t - \mu_s\|^2. \tag{10}$$

The weight $(1 - \alpha_t)$ is the leftover baseline mass.

Remark 6.1. Deriving the same term from a KL divergence toward the baseline Gaussian changes nothing: the μ_s-dependent content of any such divergence is exactly this weighted MSE, because a fixed-variance Gaussian puts its mean only in the squared distance. The isotropic choice is what makes the distance Euclidean.

6.3 The M-Step over the Subject Mean

Proposition 6.2. *With α fixed, the M-step over μ_s gives:*

$$\mu_s^\star = \frac{\sum_{t \in s} (1 - \alpha_t)\, z_t}{\sum_{t \in s} (1 - \alpha_t)}.$$

Proof. Set $\nabla_{\mu_s} \mathcal{L}_{\text{sub}} = 0$. Using $\nabla_\mu \|z - \mu\|^2 = -2(z - \mu)$:

$$\sum_{t \in s} (1 - \alpha_t)(z_t - \mu_s) = 0 \implies \mu_s \sum_{t \in s} (1 - \alpha_t) = \sum_{t \in s} (1 - \alpha_t) z_t.$$

By Remark 4.2, the effective baseline mass $N_s = \sum_{t \in s} (1 - \alpha_t) \geq T_s (1 - \alpha_{\max}) > 0$ is strictly positive, so dividing by N_s is always well-defined. □

6.4 The Low-Rank Model and Global Centering

Define the global baseline mean $\mu = \frac{\sum_s N_s \mu_s^\star}{\sum_s N_s}$ and centre all embeddings as $h_t = z_t - \mu$. Impose the low-rank structure on the centred subject means:

$$\mu_s - \mu = U a_s, \qquad U \in \mathbb{R}^{d \times m}, \quad m < d, \quad U^\top U = I_m.$$

Note that $UU^\top$ is a rank-m orthogonal projector, not the identity. In terms of the centred embeddings $h_t = z_t - \mu$, the subject loss (10) is $\mathcal{L}_{\text{sub}} = \sum_t (1 - \alpha_t) \|h_t - U a_{s_t}\|^2$.

6.5 The M-Step over U: Between-Subject Scatter and Refresh Cadence

The M-step over $(U, \{a_s\})$ given $\{\mu_s^\star\}$ is:

$$\min_{U,\{a_s\}} \sum_s N_s \|(\mu_s^\star - \mu) - U a_s\|^2.$$

Proposition 6.3. *The optimal U comprises the top-m eigenvectors of the between-subject scatter matrix*

$$S_{\text{between}} = \frac{1}{N} \sum_{s=1}^{S} N_s (\mu_s^\star - \mu)(\mu_s^\star - \mu)^\top, \qquad N = \sum_{s=1}^{S} N_s.$$

Proof. Write $v_s = \mu_s^\star - \mu$. For fixed U, the inner optimum over a_s is $a_s^\star = U^\top v_s$, and the residual is $\|v_s\|^2 - \|U^\top v_s\|^2$. Summing with weights N_s:

$$\sum_s N_s \|v_s\|^2 - \sum_s N_s \|U^\top v_s\|^2.$$

The first term does not depend on U, so minimising the residual is maximising:

$$\sum_s N_s \|U^\top v_s\|^2 = \sum_s N_s \operatorname{tr}(U^\top v_s v_s^\top U) = N \operatorname{tr}(U^\top S_{\text{between}} U),$$

where we used the trace identity $\|U^\top v\|^2 = \operatorname{tr}(U^\top v v^\top U)$. By the Rayleigh–Ritz theorem, the orthonormal U maximising this trace is the top-m eigenvectors of S_{between}. These are the directions along which subject means differ—the subject-separating directions. □

Remark 6.4 (Refresh cadence and zero-lag tracking of U). During Phase 1A training, the model maintains an $S \times d$ exponential moving average (EMA) table of subject means $\bar{\mu}_s$ and baseline masses $\bar{N}_s$ updated every mini-batch with momentum m (matching the queue momentum in Section 9.4). Because $S \leq 16$ in SEED-V, U is obtained as the top-m left singular vectors of the $d \times S$ matrix $M_{\text{sub}} = [\sqrt{\bar{N}_1/\bar{N}}(\bar{\mu}_1 - \mu), \ldots, \sqrt{\bar{N}_S/\bar{N}}(\bar{\mu}_S - \mu)] \in \mathbb{R}^{d \times S}$. A thin SVD of a 256×16 matrix requires $\mathcal{O}(dS^2)$ operations (< 0.1 ms on GPU), allowing U to be refreshed every step (or every epoch) with staleness bounded by the exact queue staleness bound of Section 9.4. At the conclusion of Phase 1A, one full-pass M-step over the frozen training embeddings locks in the exact $(\mu, U, \{\rho_y\})$ for Phase 1B.

6.6 The Projection is the M-Step Readout (and Zero-Shot Unseen Subject Removal)

Any centred embedding $h_t = z_t - \mu$ splits orthogonally as

$$h_t = UU^\top h_t + (I - UU^\top) h_t,$$

its coordinate in $\operatorname{span}(U)$ plus the orthogonal residual. The generative model says the subject baseline mean lives in $\mu + \operatorname{span}(U)$, so all subject-specific variation is carried by the $\operatorname{span}(U)$ coordinates. Therefore removing them with

$$h_t' = (I - UU^\top) h_t = (I - UU^\top)(z_t - \mu) \tag{11}$$

is not an ad hoc trick: it is the exact residual the subject branch cannot explain, which is where the emotion signal lives.

Remark 6.5 (Zero-shot generalisation to unseen test subjects). Crucially, for any unseen test subject s_{test} whose baseline mean lies in the low-rank subspace $\mu_{s_{\text{test}}} = \mu + U a_{s_{\text{test}}}$ with unknown coordinates $a_{s_{\text{test}}} \in \mathbb{R}^m$, the projector annihilates the subject offset **without needing to estimate** $a_{s_{\text{test}}}$:

$$(I - UU^\top)(z_t - \mu_{s_{\text{test}}}) = (I - UU^\top)(z_t - \mu) - \underbrace{(I - UU^\top)U}_{=0} a_{s_{\text{test}}} = h_t'.$$

Remark 6.6 (Rank constraint). If $m = d$, then U is a full orthogonal basis, $UU^\top = I$, the nullspace is $\{0\}$, and the projection annihilates everything. Structurally, the centred vectors $(\mu_s^\star - \mu)$ satisfy $\sum_s N_s(\mu_s^\star - \mu) = 0$, so they span at most $S-1$ directions and $\operatorname{rank}(S_{\text{between}}) \le S-1$. Practically, $m \le S-1$ is chosen at the spectral elbow and validated by the linear subject probe.

7 The Informativeness Anchor (Pure Projected Anchor)

7.1 Mutual Information of the Subject-Free Representation as the Target

The ELBO needs one term preventing collapse in the exact subject-free space $\operatorname{span}(U)^\perp$ used by OT and the Phase 1B readout. We therefore certify the **pure projected anchor** (keeping a 50/50 split with raw z_t strictly as an empirical ablation row):

$$I(x;\tilde{z}) = \mathbb{E}_{p(x,\tilde{z})}\left[\log \frac{p(x,\tilde{z})}{p(x)\,p(\tilde{z})}\right].$$

Define the density ratio

$$r(x,\tilde{z}) = \frac{p(x,\tilde{z})}{p(x)p(\tilde{z})} = \frac{p(\tilde{z}\mid x)}{p(\tilde{z})}.$$

7.2 Optimal Real-vs-Random Score

Proposition 7.1. *The optimal score for the binary classification problem $(x,\tilde{z}) \sim p(x,\tilde{z})$ (real) vs. $(x,\tilde{z}) \sim p(x)p(\tilde{z})$ (random) is $s^\star(x,\tilde{z}) = \log r(x,\tilde{z})$, the pointwise mutual information.*

Proof. The population binary loss for a free scalar score $s(x,\tilde{z})$ is:

$$\mathcal{L}_{\text{bin}}[s] = -\mathbb{E}_{p(x,\tilde{z})}[\log\sigma(s(x,\tilde{z}))] - \mathbb{E}_{p(x)p(\tilde{z})}[\log(1-\sigma(s(x,\tilde{z})))].$$

Since the loss depends on s pointwise, fix one pair $(x,\tilde{z})$ and write $a = p(x,\tilde{z})$, $b = p(x)p(\tilde{z})$. The local scalar objective is $g(v) = -a\log\sigma(v) - b\log(1-\sigma(v))$. Using $\frac{d}{dv}\log\sigma(v) = 1-\sigma(v)$ and $\frac{d}{dv}\log(1-\sigma(v)) = -\sigma(v)$, the stationary condition is $-a(1-\sigma(v^\star)) + b\sigma(v^\star) = 0$. Thus $\sigma(v^\star) = a/(a+b)$, and inverting the sigmoid gives

$$s^\star(x,\tilde{z}) = \log\frac{a}{b} = \log\frac{p(x,\tilde{z})}{p(x)p(\tilde{z})} = \log r(x,\tilde{z}).$$

Averaging under the joint recovers mutual information: $\mathbb{E}_{p(x,\tilde{z})}[s^\star(x,\tilde{z})] = I(x;\tilde{z})$. □

7.3 The Coupled InfoNCE Certificate

Theorem 7.2 (Coupled InfoNCE Bound)**.** *For the K-way game with one positive $\tilde{z}_1 \sim p(\tilde{z}\mid x)$ and $K-1$ negatives $\tilde{z}_k \sim p(\tilde{z})$:*

$$\boxed{I(x;\tilde{z}) \ge \log K - \mathcal{L}_{\text{NCE}}[s]}$$

for any score function s.

Proof. The coupled InfoNCE loss [6] is $\mathcal{L}_{\text{NCE}}[s] = \mathbb{E}\left[-\log\frac{e^{s(x,\tilde{z}_1)}}{\sum_{k=1}^K e^{s(x,\tilde{z}_k)}}\right]$. The optimal softmax classifier is the true posterior over the real index. Comparing slot 1 to slot k, Bayes gives:

$$\frac{P(1 \text{ real}\mid x,\tilde{z}_{1:K})}{P(k \text{ real}\mid x,\tilde{z}_{1:K})} = \frac{p(\tilde{z}_1\mid x)\,p(\tilde{z}_k)}{p(\tilde{z}_k\mid x)\,p(\tilde{z}_1)} = \frac{r(x,\tilde{z}_1)}{r(x,\tilde{z}_k)}.$$

Therefore the optimal score satisfies $s^\star(x,\tilde{z}) = \log r(x,\tilde{z}) + c(x)$, where $c(x)$ cancels in the softmax. Substituting $e^{s^\star} \propto r$:

$$\mathcal{L}^\star_{\mathrm{NCE}} = \mathbb{E}\left[-\log \frac{r_1}{\sum_{k=1}^K r_k}\right] = \mathbb{E}\left[\log \sum_{k=1}^K r_k\right] - \mathbb{E}[\log r_1].$$

Since $\tilde{z}_1 \sim p(\tilde{z} \mid x)$, we have $\mathbb{E}[\log r_1] = I(x;\tilde{z})$. Writing $\log \sum_{k=1}^K r_k = \log K + \log\Big(\frac{1}{K}\sum_{k=1}^K r_k\Big)$ gives $\log K - \mathcal{L}^\star_{\mathrm{NCE}} = I(x;\tilde{z}) - C_K$, where $C_K = \mathbb{E}[\log(\frac{1}{K}\sum_k r_k)]$. Let $T = \frac{1}{K}\sum_k r_k$. For $T > 0$, $\log T \geq 1 - 1/T$, so

$$C_K \geq 1 - \mathbb{E}\left[\frac{1}{T}\right] = 1 - \mathbb{E}\left[\frac{K}{\sum_k r_k}\right].$$

Under the product measure $\prod_j p(\tilde{z}_j)$, the variables $\tilde{z}_1, \ldots, \tilde{z}_K$ are exchangeable and $\sum_k r_k$ is symmetric. Averaging over $m = 1, \ldots, K$:

$$\int \frac{K r_1}{\sum_k r_k} p(x) \prod_j p(\tilde{z}_j)\, dx\, d\tilde{z}_{1:K} = \int \frac{\sum_m r_m}{\sum_k r_k} p(x) \prod_j p(\tilde{z}_j)\, dx\, d\tilde{z}_{1:K} = 1.$$

Thus $C_K \geq 0$. Since $s^\star$ minimises the loss, any score s has $\mathcal{L}_{\mathrm{NCE}}[s] \geq \mathcal{L}^\star_{\mathrm{NCE}}$, giving $I(x;\tilde{z}) \geq \log K - \mathcal{L}_{\mathrm{NCE}}[s]$. □

7.4 The Decoupled (DCL) Form

The implementation removes the positive from the negative partition [7]. With $K-1$ negatives, define $Y_x = \frac{1}{K-1}\sum_{k=2}^K e^{s(x,\tilde{z}_k)}$. The population DCL loss is:

$$\mathcal{L}_{\mathrm{DCL}}[s] = -\mathbb{E}_{p(x,\tilde{z})}[s(x,\tilde{z})] + \mathbb{E}_{p(x)\prod_{k=2}^K p(\tilde{z}_k)}\left[\log \sum_{k=2}^K e^{s(x,\tilde{z}_k)}\right].$$

Using $\log \sum_{k=2}^K e^{s(x,\tilde{z}_k)} = \log(K-1) + \log Y_x$, define the finite negative certificate:

$$C_K^{\mathrm{DCL}}(s) = \log(K-1) - \mathcal{L}_{\mathrm{DCL}}[s].$$

Theorem 7.3 (DCL Information Bound)**.** *In the population limit,*

$$I_{\mathrm{DV}}(s) \leq C_\infty^{\mathrm{DCL}}(s) \leq I(x;\tilde{z}),$$

where I_{DV} is the Donsker–Varadhan bound.

Proof. We have $C_K^{\mathrm{DCL}}(s) = \mathbb{E}_{p(x,\tilde{z})}[s(x,\tilde{z})] - \mathbb{E}_{x,\mathrm{neg}}[\log Y_x]$. For fixed x, define $h(x) = \mathbb{E}_{p(\tilde{z})}[e^{s(x,\tilde{z})}]$. Since negatives are i.i.d. from $p(\tilde{z})$, $\mathbb{E}_{\mathrm{neg}}[Y_x] = h(x)$. By concavity of log, $\mathbb{E}_{\mathrm{neg}}[\log Y_x] \leq \log h(x)$, so $C_K^{\mathrm{DCL}}(s) \geq C_\infty^{\mathrm{DCL}}(s) := \mathbb{E}_{p(x,\tilde{z})}[s(x,\tilde{z})] - \mathbb{E}_{p(x)}[\log h(x)]$. Next, for fixed x, let $X_{\tilde{z}} = e^{s(x,\tilde{z})}/r(x,\tilde{z})$. Using $p(\tilde{z} \mid x) = r(x,\tilde{z})p(\tilde{z})$, $\mathbb{E}_{p(\tilde{z}|x)}[X_{\tilde{z}}] = h(x)$. By Jensen's inequality on $-\log$, $\mathbb{E}_{p(\tilde{z}|x)}[\log r(x,\tilde{z}) - s(x,\tilde{z})] \geq -\log h(x)$. Integrating over $p(x)$ yields $I(x;\tilde{z}) - C_\infty^{\mathrm{DCL}}(s) \geq 0$. Finally, applying Jensen to $\mathbb{E}_{p(x)}[\log h(x)] \leq \log \mathbb{E}_{p(x)p(\tilde{z})}[e^{s(x,\tilde{z})}]$ gives $I_{\mathrm{DV}}(s) \leq C_\infty^{\mathrm{DCL}}(s)$. □

Remark 7.4 (Role in the ELBO: Pure Projected Anchor Certifies $\mathrm{span}(U)^\perp$). Evaluating $\mathcal{L}_{\mathrm{anc}}$ directly on the projected representation $\tilde{z}_t \in \mathbb{B}^d \cap \mathrm{span}(U)^\perp$ guarantees that mutual information $I(x;\tilde{z})$ is maintained inside the exact nullspace used by OT and the Phase 1B readout R_b, preventing the encoder from hiding all variance inside $\mathrm{span}(U)$.

8 Gradient Analysis and Coefficient Design on the Closed Unit Ball $\mathbb{B}^d$

8.1 Per-Query Gradient Bound on $\|\tilde{z}_i\| \le 1$

Proposition 8.1. *On the closed unit ball domain $\|\tilde{z}_i\| \le 1$, $\|r_j\| \le 1$ under stop-gradient coupling:*

(a) For the cross-matching functional $W_\varepsilon^y(\mu_y, \nu_y)$ (or one-sided queue-debiased $W_\varepsilon^y(\mu_y, \nu_y) - W_\varepsilon^y(\nu_y, \nu_y)$), the per-query gradient with respect to $\tilde{z}_i$ satisfies

$$\|\nabla_{\tilde{z}_i} W_\varepsilon^y(\mu_y, \nu_y)\| \le \sum_{j \in Q_y} \Pi_{ij} \le a_i \le 1.$$

(b) For the symmetric debiased divergence $S_\varepsilon^y(\mu_y, \nu_y) = W_\varepsilon^y(\mu_y, \nu_y) - \frac{1}{2} W_\varepsilon^y(\mu_y, \mu_y) - \frac{1}{2} W_\varepsilon^y(\nu_y, \nu_y)$ with optimal self-coupling Π^{self}, the gradient satisfies

$$\|\nabla_{\tilde{z}_i} S_\varepsilon^y(\mu_y, \nu_y)\| \le 2a_i \le 2.$$

Proof. (a) The coupling is stop-gradient (justified by Proposition 8.3), so gradient flows only through the cost $C_{ij} = 1 - \tilde{z}_i^\top r_j$, giving $\nabla_{\tilde{z}_i} C_{ij} = -r_j$. Therefore:

$$\nabla_{\tilde{z}_i} W_\varepsilon^y(\mu_y, \nu_y) = -\sum_{j \in Q_y} \Pi_{ij} r_j.$$

By the triangle inequality and $\|r_j\| \le 1$ on $\mathbb{B}^d$:

$$\left\| \sum_{j \in Q_y} \Pi_{ij} r_j \right\| \le \sum_{j \in Q_y} \Pi_{ij} \|r_j\| \le \sum_{j \in Q_y} \Pi_{ij} = a_i - d_i \le a_i \le 1.$$

(b) In $\frac{1}{2} W_\varepsilon^y(\mu_y, \mu_y)$, $\tilde{z}_i$ appears in both row i and column i of $C_{ik}^{\text{self}} = 1 - \tilde{z}_i^\top \tilde{z}_k$. By symmetry of Π^{self}, $\nabla_{\tilde{z}_i} \left[\frac{1}{2} W_\varepsilon^y(\mu_y, \mu_y) \right] = -\sum_{k \in I_y} \Pi_{ik}^{\text{self}} \tilde{z}_k$, whose norm is bounded by $\sum_k \Pi_{ik}^{\text{self}} \|\tilde{z}_k\| \le a_i$. Combining with (a) via the triangle inequality gives $\|\nabla_{\tilde{z}_i} S_\varepsilon^y\| \le 2a_i \le 2$. □

8.2 Two Zero-Gradient Regimes (Anti-Ghost Mechanism)

Proposition 8.2. *The OT cross-matching gradient vanishes in two regimes on $\mathbb{B}^d$:*

(a) ***Fully baseline window.*** *$\alpha_i \to 0 \implies a_i \to 0 \implies$ row mass vanishes $\implies$ gradient is zero across all of $\mathbb{B}^d$.*

(b) ***Fully confident match.*** *One-hot Π and $\tilde{z}_i^\top r_{j^\star} \to 1$. Since $\|\tilde{z}_i\| \le 1$ and $\|r_{j^\star}\| \le 1$, $\tilde{z}_i^\top r_{j^\star} \to 1$ implies $\|\tilde{z}_i\| \to 1$ (active spherical regime $\|h_i'\| \ge \tau_{\text{norm}}$), where the radial gradient is annihilated by normalisation and the tangential gradient has norm $a_i \sqrt{1 - (\tilde{z}_i^\top r_{j^\star})^2} \to 0$.*

Proof. (a) If $\alpha_i \to 0$, then $a_i = \alpha_i / A_y \to 0$. The row mass $\sum_j \Pi_{ij} \le a_i \to 0$, forcing $\|\nabla_{\tilde{z}_i} \mathcal{L}_{\text{em}}\| \to 0$. The OT term cannot manufacture a burst out of baseline.

(b) By Cauchy–Schwarz on $\mathbb{B}^d$, $\tilde{z}_i^\top r_{j^\star} \le \|\tilde{z}_i\| \|r_{j^\star}\| \le 1$, so $\tilde{z}_i^\top r_{j^\star} \to 1$ forces $\|\tilde{z}_i\| = 1$ (meaning $\|h_i'\| \ge \tau_{\text{norm}}$ and $\tilde{z}_i = h_i' / \|h_i'\|$). With one-hot Π putting all mass on $r_{j^\star}$, the gradient in $\tilde{z}_i$ is $-a_i r_{j^\star}$. Back-propagating through $\tilde{z}_i = h_i' / \|h_i'\|$ applies $(I - \tilde{z}_i \tilde{z}_i^\top) / \|h_i'\|$, which cancels the radial component $\tilde{z}_i$ and leaves tangential norm $\frac{a_i}{\|h_i'\|} \sqrt{\|r_{j^\star}\|^2 - (\tilde{z}_i^\top r_{j^\star})^2} \le \frac{a_i}{\tau_{\text{norm}}} \sqrt{1 - (\tilde{z}_i^\top r_{j^\star})^2} \to 0$ as $\tilde{z}_i^\top r_{j^\star} \to 1$. A settled true burst stops moving. No ghost drift. □

8.3 The Gradient-Norm Ratio and λ_{OT}

Because both the pure projected anchor $\mathcal{L}_{\mathrm{anc}}(\tilde{z}_t)$ and the emotion OT term $\mathcal{L}_{\mathrm{em}}(\tilde{z}_t)$ act on the exact same variable $\tilde{z}_t \in \mathbb{B}^d \cap \mathrm{span}(U)^{\perp}$, we scale the OT term by λ_{OT} and require:

$$\lambda_{\mathrm{OT}} \|\nabla_{\tilde{z}} \mathcal{L}_{\mathrm{em}}\| \le c \|\nabla_{\tilde{z}} \mathcal{L}_{\mathrm{anc}}\| \implies \lambda_{\mathrm{OT}}^{\max} = c \frac{\|\nabla_{\tilde{z}} \mathcal{L}_{\mathrm{anc}}\|}{\|\nabla_{\tilde{z}} \mathcal{L}_{\mathrm{em}}\|}, \qquad c \in [c_{\min}, c_{\max}].$$

Since both norms are $\mathcal{O}(1)$ per query, $\lambda_{\mathrm{OT}}^{\max}$ is $\mathcal{O}(1) \times c$.

- **Upper end $c_{\max} < 1$:** the largest c such that the combined gradient $\nabla_{\tilde{z}}(\mathcal{L}_{\mathrm{anc}} + \lambda_{\mathrm{OT}} \mathcal{L}_{\mathrm{em}})$ maintains a strictly negative directional derivative along the anchor's escape direction away from constant collapse.
- **Lower end $c_{\min} > 0$:** the smallest c such that the OT term still moves burst-flagged windows.

[Track B Proof Guide (for Nur) — B5(iii) — 1D Directional Derivative Proof for $c_{\max} < 1$.
Step 1 (1D line model): Parameterise a straight line moving away from collapse by a single scalar $t \ge 0$, and write the loss along that line as $\phi(t) = f_{\mathrm{anc}}(t) + \lambda_{\mathrm{OT}} f_{\mathrm{em}}(t)$, where the anchor decreases the loss at rate $f'_{\mathrm{anc}}(0) = -A < 0$ ($A > 0$) and the OT slope has magnitude $|f'_{\mathrm{em}}(0)| \le E$. Substitute $\lambda_{\mathrm{OT}} = c\, g\, \frac{A}{E}$ with $g \in [0,1]$ into the 1D derivative at $t = 0$:

$$\phi'(0) = f'_{\mathrm{anc}}(0) + \lambda_{\mathrm{OT}} f'_{\mathrm{em}}(0) \le -A + \left(c\, g\, \frac{A}{E}\right) E = -(1 - c\, g) A.$$

Since $g \in [0,1]$ and $A > 0$, $\phi'(0) < 0$ is guaranteed for all $g \in [0,1]$ if and only if $1 - c > 0 \iff c \le c_{\max} < 1$.
Step 2 (Extension to $\mathbb{B}^d$): Set $A = \|\nabla_{\tilde{z}} \mathcal{L}_{\mathrm{anc}}\|$ and $E = \|\nabla_{\tilde{z}} \mathcal{L}_{\mathrm{em}}\|$ along the unit escape direction $u = -\nabla_{\tilde{z}} \mathcal{L}_{\mathrm{anc}} / A$; the basic dot-product inequality $v \cdot u \le \|v\|\|u\| = \|v\|$ reduces the d-dimensional directional derivative to Step 1.]

8.4 Stop-Gradient on the Coupling is Exact

Proposition 8.3 (Envelope Theorem for OT)**.** *The regularised OT value $W(\tilde{C}) = \min_{\tilde{\Pi}} \langle \tilde{\Pi}, \tilde{C} \rangle + R_{\varepsilon}(\tilde{\Pi})$ over the compact simplex of couplings satisfies:*

$$\left.\frac{dW}{dC_{ij}}\right|_{\tilde{\Pi} = \tilde{\Pi}^{\star}} = \Pi^{\star}_{ij}.$$

Proof. Because $R_{\varepsilon}(\tilde{\Pi})$ is strictly convex and has steep boundary gradients ($\log \tilde{\Pi}_{ij} \to -\infty$ as $\tilde{\Pi}_{ij} \to 0^{+}$) on a compact polytope with strictly positive margins, the unique optimum $\tilde{\Pi}^{\star}$ lies in the strict relative interior of the polytope. At $\tilde{\Pi}^{\star}$, the KKT stationarity condition annihilates the $\partial / \partial \tilde{\Pi}$ derivative along all feasible tangent directions. By Danskin's envelope theorem:

$$\nabla_{\tilde{z}} W = \sum_{i,j \ge 1} \Pi^{\star}_{ij} \nabla_{\tilde{z}} C_{ij}.$$

The coupling's own dependence on $\tilde{z}$ contributes nothing at first order. Therefore treating $\tilde{\Pi}^{\star}$ as stop-gradient loses nothing at first order and avoids the dead-list pathology of unrolling the Sinkhorn fixed point. □

9 Queue Estimator Design

9.1 The Two Sides and the Zero-Gradient Proof

Proposition 9.1 (Queue-vs-Queue is Dead)**.** *If both sides of the OT match are stop-gradient (queue-vs-queue), then* $\nabla_\theta \mathcal{L}_{\mathrm{OT}} = 0$.

Proof. If both sides are stop-gradient, the loss depends only on quantities that carry no gradient to the online parameters θ. Therefore the loss is constant in θ and its gradient vanishes. A zero-gradient term is dead weight. Therefore the OT must be current-vs-queue. □

9.2 One Queue, Two Readouts

Store one entry per window: its nullspace-projected feature $r_j \in \mathbb{B}^d$, its class y_j, its stored burst weight $\beta_j = \alpha_j^{(1\mathrm{A})}$, and its window ID. Enqueue every window, and read the queue two ways:

- **OT reference:** per-class, β-weighted on nullspace keys r_j. Converges to the emotion prototype p_y.
- **Anchor negatives:** whole queue, uniform on nullspace keys r_j. Converges to the nullspace marginal $p(\tilde{z})$.

9.3 Convergence of the β-Weighted Readout

Proposition 9.2. *For any measurable region* $R \subset \mathbb{B}^d$*:*

$$\nu_y(R) = \frac{\sum_{j:r_j \in R} \beta_j}{\sum_j \beta_j} \xrightarrow{N_q \to \infty} \int_R p_y(\tilde{z})\, d\tilde{z}.$$

Proof. Class-y entries in the queue are samples from the class mixture $p(\tilde{z} \mid y) = \rho_y p_y(\tilde{z}) + (1 - \rho_y) p_{\mathrm{base}}(\tilde{z})$, and each carries weight $\beta_j = p(e = 1 \mid \tilde{z}_j, y)$. By the law of large numbers, as the queue grows:

$$\nu_y(R) \longrightarrow \frac{\int_R \beta(\tilde{z})\, p(\tilde{z} \mid y)\, d\tilde{z}}{\int \beta(\tilde{z})\, p(\tilde{z} \mid y)\, d\tilde{z}}.$$

In the numerator, by Bayes' rule: $\beta(\tilde{z}) p(\tilde{z} \mid y) = p(e = 1 \mid \tilde{z}, y) p(\tilde{z} \mid y) = p(e = 1, \tilde{z} \mid y) = \rho_y p_y(\tilde{z})$. In the denominator: $\int \beta(\tilde{z}) p(\tilde{z} \mid y)\, d\tilde{z} = p(e = 1 \mid y) = \rho_y$. Therefore:

$$\nu_y(R) \longrightarrow \frac{\rho_y \int_R p_y(\tilde{z})\, d\tilde{z}}{\rho_y} = \int_R p_y(\tilde{z})\, d\tilde{z}.$$

The factor ρ_y cancels, and the baseline component disappears. □

9.4 Leakage Guard and Staleness Bound

Per-sample masking (store window ID, mask same-window keys at read time) replaces the risky "soft-ignore stale points" argument. Staleness is bounded:

$$\text{staleness} \le L \frac{N_q}{B}(1 - m)\, D, \qquad \frac{N_q}{B}(1 - m) \le \frac{\varepsilon_{\mathrm{tol}}}{LD}.$$

9.5 The Burst Prior ρ_y: Seeding and Refinement

By the law of total expectation, $\rho_y = P(e = 1 \mid y) = \mathbb{E}_t[\alpha_t \mid y]$. The M-step of the sparsity term gives the self-consistency equation:

$$\rho_y = \frac{1}{N_y} \sum_{t \in y} \alpha_t, \tag{12}$$

clamped to $[\rho_{\min}, 1 - \rho_{\min}]$ to preserve strict interiority.

Seed (Option B: threshold-free fixed point). After CWT, log1p, and per-subject robust standardisation, the baseline bulk is $\approx \mathcal{N}(0, I)$. Define $d_t^2 = \|r_t\|^2$ and the log-likelihood ratio:

$$\Delta_t = \frac{1 - s^{-2}}{2} d_t^2 - B \log s,$$

with $s > 1$ the burst spread constant. The self-consistency map $F(u) = \operatorname{logit}\Big(\frac{1}{T_y} \sum_{t \in y} \sigma(\Delta_t + u)\Big)$ is a contraction.

Contraction proof. With $P_t = \sigma(\Delta_t + u)$ and $m = \frac{1}{T_y} \sum_t P_t$:

$$F'(u) = \frac{\mathbb{E}[P(1 - P)]}{m(1 - m)}.$$

Using the identity $m(1 - m) = \mathbb{E}[P(1 - P)] + \operatorname{Var}(P)$:

$$F'(u) = \frac{\mathbb{E}[P(1 - P)]}{\mathbb{E}[P(1 - P)] + \operatorname{Var}(P)} < 1$$

whenever the scores are non-degenerate ($\operatorname{Var}(P) > 0$). F maps the real line into the compact interval $[\min_t \Delta_t, \max_t \Delta_t]$, so on that invariant compact interval the continuous derivative $F'(u)$ attains a maximum $\kappa < 1$. By the Banach fixed-point theorem there is a unique fixed point and plain iteration reaches it geometrically. □

[Track B Proof Guide (for Nur) — B6(v) — 1D Calculus Proof of bkl **Minimiser and Contraction Fixed Points.**
Step 1 (bkl **in 1D): Fix** $a \in [\alpha_{\min}, \alpha_{\max}] \subset (0, 1)$ **and differentiate** $\phi(\rho) = \sum_{t=1}^{N_y} \left[\alpha_t \log(\alpha_t/\rho) + (1 - \alpha_t) \log(\frac{1-\alpha_t}{1-\rho})\right]$ **with respect to** $\rho \in (0, 1)$**:**

$$\phi'(\rho) = \sum_{t=1}^{N_y} \left(-\frac{\alpha_t}{\rho} + \frac{1 - \alpha_t}{1 - \rho} \right) = \frac{N_y}{\rho(1 - \rho)} \left(\rho - \frac{1}{N_y} \sum_{t=1}^{N_y} \alpha_t \right).$$

Setting $\phi'(\rho) = 0$ **gives the unique minimiser** $\rho_y(\alpha) = \frac{1}{N_y} \sum_{t \in y} \alpha_t$**, which is a continuous linear average on the compact box** $[\alpha_{\min}, \alpha_{\max}]^{N_y}$**.**
Step 2 (Seeding contraction $F(u)$**): Since** $F'(u)$ **is continuous on the closed bounded interval** $I = [\min_t \Delta_t, \max_t \Delta_t]$ **and satisfies** $0 < F'(u) < 1$ **for every** $u \in I$**, by the Extreme Value Theorem (Heine–Borel)** $F'(u)$ **attains its maximum** $\kappa = \max_{u \in I} F'(u) < 1$**, so** $|F(u_1) - F(u_2)| \le \kappa |u_1 - u_2|$ **by the 1D Mean Value Theorem.]**

9.6 Queue Size $N_q^\star$

Floor (prototype richness):

$$N_q \ge \frac{C \cdot d_{\text{eff}}}{\text{tol}^2 \, \tilde{\rho}_{\min}}, \qquad \tilde{\rho}_{\min} = \min_y \rho_y \cdot C\pi_y.$$

Ceiling (staleness):

$$N_q \le \frac{\varepsilon_{\text{tol}}\, B\, T_{\text{total}}}{10\, L\, D}.$$

$N_q^\star$ is the floor rounded up, accepted only if it lies under the ceiling.

9.7 The Readiness Gate

The OT term is gated by the anchor's own certificate:

$$g = \text{clamp}\left(\frac{C_K^{\text{DCL}}}{\log(K-1)},\, g_{\text{floor}},\, 1\right), \qquad \lambda_{\text{OT}} = c\,\frac{\|\nabla_{\tilde{z}}\mathcal{L}_{\text{anc}}\|}{\|\nabla_{\tilde{z}}\mathcal{L}_{\text{em}}\|}\, g. \tag{13}$$

The gate is driven by the always-trained backbone and opens if and only if the encoder becomes informative.

10 Phase 1B: The Sufficient Statistic Readout and MIL Adapter

When transitioning from Phase 1A to Phase 1B (and test-time inference on an unseen subject), the encoder f, the global baseline mean μ, the subject subspace $U \in \mathbb{R}^{d\times m}$, the $C = 5$ class prototype clouds $\{\nu_y\}_{y=1}^C$, and the burst priors $\{\rho_y\}_{y=1}^C$ are frozen. Both the Phase 1B responsibility gate $\alpha_t^{(1\text{B})}$ and the bag readout R_b are computed **purely on nullspace-projected coordinates** $h_t' = (I - UU^\top)(z_t - \mu)$, so Phase 1B never touches the subject subspace $\text{span}(U)$.

10.1 The Label-Free Nullspace Marginal E-Step $\alpha_t^{(1\text{B})}$ and Two-Step Baseline Contraction

Proposition 10.1 (Label-Free Nullspace Phase 1B Responsibility)**.** *For a bag b whose class label $y \in \{1,\dots,C\}$ is unknown with uniform prior $\pi_y = 1/C$, the exact marginal posterior responsibility $\alpha_t^{(1\text{B})} := P(e_t = 1 \mid h_t')$ evaluated purely on subject-free nullspace coordinates is:*

$$\alpha_t^{(1\text{B})} = \frac{\frac{1}{C}\sum_{y=1}^C \rho_y\, p_y(\tilde{z}_t)}{\frac{1}{C}\sum_{y=1}^C \rho_y\, p_y(\tilde{z}_t) + (1-\bar{\rho})\, p_s(h_t')} = \sigma\left(\log\left(\frac{1}{C}\sum_{y=1}^C \rho_y\, p_y(\tilde{z}_t)\right) - \log p_s(h_t') - \log(1-\bar{\rho})\right), \tag{14}$$

where $\bar{\rho} = \frac{1}{C}\sum_{y=1}^C \rho_y$ is the marginal burst prior, $\log p_y(\tilde{z}_t)$ is evaluated on $\tilde{z}_t = h_t'/\max(\|h_t'\|, \tau_{\text{norm}}) \in \mathbb{B}^d$ against the frozen class-y prototype cloud ν_y, and $\log p_s(h_t') = -\frac{\|h_t' - \hat{\mu}_b'\|^2}{2\sigma^2}$ uses the bag's nullspace baseline residual $\hat{\mu}_b' = G^2(\bar{h}_b')$ obtained by two steps of the baseline-weighted fixed-point map

$$G(\mu') := \frac{\sum_{t=1}^{T_b}(1 - \alpha_t^{(1\text{B})}(\mu'))\, h_t'}{\sum_{t=1}^{T_b}(1 - \alpha_t^{(1\text{B})}(\mu'))}, \qquad \mu'^{(0)} = \bar{h}_b' := \frac{1}{T_b}\sum_{t=1}^{T_b} h_t'. \tag{15}$$

Proof. By the law of total probability over $y \in \{1,\dots,C\}$ with $P(y) = 1/C$:

$$P(e_t = 1, h_t') = \sum_{y=1}^C P(y)\, P(e_t = 1 \mid y)\, p(h_t' \mid e_t = 1, y) = \frac{1}{C}\sum_{y=1}^C \rho_y\, p_y(\tilde{z}_t),$$

and since $(I - UU^\top)$ annihilates the in-span subject offset Ua_s, the nullspace baseline density $p(h_t' \mid e_t = 0, y, s) = p_s(h_t')$ carries neither an emotion index y nor an in-span subject component:

$$P(e_t = 0, h_t') = \sum_{y=1}^C P(y)(1-\rho_y)\, p_s(h_t') = (1-\bar{\rho})\, p_s(h_t').$$

Applying Bayes' rule $P(e_t = 1 \mid h'_t) = \frac{P(e_t=1,h'_t)}{P(e_t=1,h'_t)+P(e_t=0,h'_t)}$ and dividing numerator and denominator by $(1-\bar{\rho})p_s(h'_t)$ gives (14). □

Proposition 10.2 (Contraction Certificate and Error Bound for the Two-Step Baseline Estimator $\hat{\mu}'_b$)**.** *Let* $\tilde{w}_t(\mu') = \frac{1-\alpha_t^{(1B)}(\mu')}{\sum_{s=1}^{T_b}(1-\alpha_s^{(1B)}(\mu'))}$ *be the normalised baseline weights and define the burst-weighted local spread ratio*

$$\kappa_\mu := \max_{\mu'} \left[\frac{1}{\sigma^2} \sum_{t=1}^{T_b} \tilde{w}_t(\mu')\, \alpha_t^{(1B)}(\mu')\, \|h'_t - G(\mu')\|\, \|h'_t - \mu'\| \right].$$

Whenever $\kappa_\mu < 1$ *(which holds whenever the burst fraction is below one half and the baseline cluster spread is bounded by* σ^2*),* G *is a strict contraction with unique fixed point* $\mu'^\star = G(\mu'^\star)$*, and the two-step estimator* $\hat{\mu}'_b = G^2(\bar{h}'_b)$ *satisfies the explicit error bound:*

$$\|\hat{\mu}'_b - \mu'^\star\| \le \kappa_\mu^2 \|\bar{h}'_b - \mu'^\star\| \le \frac{\kappa_\mu^2}{1-\kappa_\mu} \|G(\bar{h}'_b) - \bar{h}'_b\|. \tag{16}$$

[Track B Proof Guide (for Nur) — B6(vi) — 1D Proof of the Two-Step Baseline Contraction Bound (16).
Step 1 (1D scalar case): Let $h_1, \dots, h_T \in [-1,1] \subset \mathbb{R}$ **be scalars and** $G(\mu) = \frac{\sum_t w_t(\mu) h_t}{\sum_t w_t(\mu)}$**, where** $w_t(\mu) = 1 - \alpha_t(\mu) = \sigma(-\frac{(h_t-\mu)^2}{2\sigma^2} - c_t)$**. Using the 1D chain rule** $\sigma'(v) = \sigma(v)(1-\sigma(v)) = w_t\alpha_t$**, compute** $w'_t(\mu) = w_t\alpha_t\frac{h_t-\mu}{\sigma^2}$**. Apply the 1D quotient rule** $(U/V)' = \frac{U'}{V} - \frac{U}{V}\frac{V'}{V}$ **to obtain:**

$$G'(\mu) = \frac{1}{\sigma^2}\sum_{t=1}^{T} \tilde{w}_t\alpha_t(h_t-\mu)h_t - G(\mu)\frac{1}{\sigma^2}\sum_{t=1}^{T}\tilde{w}_t\alpha_t(h_t-\mu) = \frac{1}{\sigma^2}\sum_{t=1}^{T}\tilde{w}_t\alpha_t(h_t - G(\mu))(h_t-\mu).$$

By the triangle inequality, $|G'(\mu)| \le \kappa_\mu < 1$**. By the 1D Mean Value Theorem,** $|G(x) - G(y)| \le \kappa_\mu|x-y|$**. Iterating twice from** $\mu^{(0)} = \bar{h}_b$ **gives** $|G^2(\bar{h}_b) - \mu^\star| \le \kappa_\mu|G(\bar{h}_b) - \mu^\star| \le \kappa_\mu^2|\bar{h}_b - \mu^\star|$**, and using** $|\bar{h}_b - \mu^\star| \le |\bar{h}_b - G(\bar{h}_b)| + |G(\bar{h}_b) - G(\mu^\star)|$ **gives** $|\bar{h}_b - \mu^\star| \le \frac{1}{1-\kappa_\mu}|G(\bar{h}_b) - \bar{h}_b|$**.**
Step 2 (Extension to $\mathbb{R}^d$**): Restrict** G **to the 1D line segment** $\mu'(s) = (1-s)\mu'_1 + s\mu'_2$ **for** $s \in [0,1]$ **and apply Step 1.]**

10.2 Length-Normalised Sufficient Statistic Readout R_b

Proposition 10.3 (Phase 1B Readout)**.** *The trial-length-invariant classification readout for bag* b *is the window-averaged, unnormalised-by-*α *posterior-weighted sum in the subject nullspace:*

$$R_b = \frac{1}{T_b}\sum_{t=1}^{T_b} \alpha_t^{(1B)} h'_t, \qquad h'_t = (I - UU^\top)(z_t - \mu). \tag{17}$$

Justification. Under the generative model, burst windows of class y have expectation $\mu_y = \mathbb{E}[h'_t \mid e_t = 1, y]$ and occur with probability ρ_y, while baseline windows lie in $\mu + \mathrm{span}(U)$ and are annihilated by $(I - UU^\top)$. Therefore, in expectation for any trial length T_b:

$$\mathbb{E}[R_b \mid y] = \frac{1}{T_b}\sum_{t=1}^{T_b}\mathbb{E}[e_t h'_t \mid y] = \rho_y\mu_y.$$

Dividing by the known window count T_b removes arbitrary video-duration bias across SEED-V trials while preserving the empirical burst rate $\hat{\rho}_b = \frac{1}{T_b}\sum_{t=1}^{T_b}\alpha_t^{(1B)} \approx \rho_y$ in the magnitude $\|R_b\|$. Direction

encodes emotion content μ_y; magnitude encodes burst rate ρ_y; and $\|R_b\| \le \frac{1}{T_b}\sum_{t=1}^{T_b} |\alpha_t^{(1B)}|\,\|h'_t\| \le 2$ is uniformly bounded independent of T_b. □

Proposition 10.4 (Attention is the Normalised Content Part; Bayes Plug-In Linear Head)**.** *The M-step for the emotion prototype direction is the normalised attention average* $\mu_y = \frac{\sum_t \alpha_t h'_t}{\sum_t \alpha_t} = \mathbb{E}_q[h']$. *Normalising by* $\sum_t \alpha_t$ *divides out the class-discriminative burst rate* $\hat{\rho}_b = \frac{1}{T_b}\sum_t \alpha_t \approx \rho_y$. *Keeping the unnormalised product*

$$R_b = \hat{\rho}_b\, \mathbb{E}_q[h']$$

couples content and burst rate in a single vector $R_b \in \mathbb{R}^d$. *Under an isotropic Gaussian bag-readout model* $R_b \mid y \sim \mathcal{N}(\rho_y\mu_y, \sigma_R^2 I)$, *the exact Bayes log-posterior classifier is a* ***single linear layer with bias*** *(`nn.Linear(d, C, bias=True)`):*

$$\ell_y(R_b) = w_y^\top R_b + b_y, \qquad w_y = \frac{\rho_y\mu_y}{\sigma_R^2}, \quad b_y = -\frac{\|\rho_y\mu_y\|^2}{2\sigma_R^2} + \log\pi_y. \tag{18}$$

Proof. Expanding the Gaussian log-likelihood $-\frac{1}{2\sigma_R^2}\|R_b - \rho_y\mu_y\|^2 + \log\pi_y = -\frac{\|R_b\|^2}{2\sigma_R^2} + \frac{(\rho_y\mu_y)^\top R_b}{\sigma_R^2} - \frac{\|\rho_y\mu_y\|^2}{2\sigma_R^2} + \log\pi_y$ and dropping the class-independent quadratic term $-\|R_b\|^2/(2\sigma_R^2)$ yields the affine logit (18). Notice that a cosine-similarity head would normalise $R_b/\|R_b\|$ and discard $\hat{\rho}_b$, whereas the linear head with bias b_y uses both the direction $\mu_y^\top R_b$ and the magnitude $\|R_b\| \approx \rho_y$. □

Remark 10.5. No temperature ensemble is needed. Reading a soft pool at several temperatures is a heuristic discretisation of one object. The single sufficient statistic R_b evaluated with a single linear layer $\ell_y(R_b) = w_y^\top R_b + b_y$ already couples content and burst rate.

11 The Assembled Objective

Minimising the negative ELBO (5) ($\mathcal{L}_{\text{total}} \equiv -\text{ELBO}$) by substituting the concrete forms from Sections 5–7 gives:

$$\mathcal{L}_{\text{em}} = \sum_{y=1}^{C} S_\varepsilon^{\text{unbal}}(\mu_y^{\text{cur}},\, \nu_y^{\text{queue}}), \tag{19}$$

$$\mathcal{L}_{\text{sub}} = \sum_t (1-\alpha_t)\,\|(z_t - \mu) - U a_{s_t}\|^2, \tag{20}$$

$$\mathcal{L}_{\text{spr}} = \sum_t \text{bkl}(\alpha_t, \rho_{y_t}), \qquad \rho_y = \frac{1}{N_y}\sum_{t\in y}\alpha_t, \tag{21}$$

$$\mathcal{L}_{\text{anc}} = \mathcal{L}_{\text{DCL}}(\tilde{z}), \qquad I_{\text{DV}}(\tilde{z}) \le C_\infty^{\text{DCL}}(\tilde{z}) \le I(x;\tilde{z}). \tag{22}$$

$$\boxed{\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{anc}} + \lambda_{\text{OT}}\,\mathcal{L}_{\text{em}} + \lambda_{\text{sub}}\,\mathcal{L}_{\text{sub}} + \lambda_\rho\,\mathcal{L}_{\text{spr}}} \tag{23}$$

with the readiness gate (13) and coefficient from Section 8.

11.1 The Gate (Restated)

$$\lambda_{\text{OT}} = c\,\frac{\|\nabla_{\tilde{z}}\mathcal{L}_{\text{anc}}\|}{\|\nabla_{\tilde{z}}\mathcal{L}_{\text{em}}\|}\,g, \qquad g = \text{clamp}\left(\frac{C_K^{\text{DCL}}}{\log(K-1)},\, g_{\text{floor}},\, 1\right), \qquad c \in [c_{\min}, c_{\max}].$$

11.2 The Phase 1B Readout and Classifier (Restated)

$$R_b = \frac{1}{T_b}\sum_{t=1}^{T_b} \alpha_t^{(1\mathrm{B})} h'_t, \qquad h'_t = (I - UU^\top)(z_t - \mu), \qquad \hat{y}_b = \arg\max_{y\in\{1,\dots,C\}} (w_y^\top R_b + b_y).$$

Direction is emotion content; magnitude is burst rate; subject-free by construction.

11.3 Why Memorisation is Impossible by Construction

(a) Both p_y and $\alpha_t^{(1\mathrm{B})}$ depend only on nullspace coordinates $h'_t \in \mathrm{span}(U)^\perp$ and carry no subject index $\implies$ the only way to reduce $\mathcal{L}_{\mathrm{em}}$ is cross-subject burst alignment.

(b) Subject baseline signal is pulled into the low-rank subspace $\mu + \mathrm{span}(U)$ by $\mathcal{L}_{\mathrm{sub}}$ and annihilated by $(I - UU^\top)$ for both seen and unseen subjects.

(c) Baselines are skipped: e_t routes them to the subject branch; the sparsity prior and drop price $\tau_y = -\varepsilon \log(1 - \rho_y)$ make Fear/Disgust prefer "mostly baseline."

11.4 Practical Engineering Guards (Preventing Secondary Instabilities)

To ensure training stability across all edge cases without introducing extra tuned modules, three implementation guards are enforced:

(i) **Pure Projected Anchor and Closed-Ball Norm Floor:** Evaluating $\mathcal{L}_{\mathrm{anc}}$ directly on $\tilde{z}_t = h'_t / \max(\|h'_t\|, \tau_{\mathrm{norm}}) \in \mathbb{B}^d$ ($\tau_{\mathrm{norm}} = 0.25$) certifies informativeness of the exact nullspace used by OT and R_b, while bounding the normalisation Jacobian by $1/\tau_{\mathrm{norm}} = 4$ during early warmup (a 50/50 split between z_t and $\tilde{z}_t$ is reserved strictly as an empirical ablation row).

(ii) **Dynamic Spectral Rank m:** For S training subjects, $\mathrm{rank}(S_{\mathrm{between}}) \le S-1$, but the smallest trailing eigenvalues reflect finite-batch sampling noise. Rather than hardcoding $m = S - 1$, m is selected dynamically as the smallest integer capturing 85%–90% of $\mathrm{tr}(S_{\mathrm{between}})$ (typically $m \in [6, 10]$ for $S = 15$), with $U = 0$ held fixed during the first 1–2 warmup epochs while the queue populates.

(iii) **Certified Two-Step Nullspace Baseline $\hat{\mu}'_b$ at Test Time:** In Phase 1B / test-time evaluation, $\hat{\mu}'_b = G^2(\bar{h}'_b)$ in $\log p_s(h'_t) = -\|h'_t - \hat{\mu}'_b\|^2/(2\sigma^2)$ is computed in nullspace coordinates via two steps of the contraction map (15) (or across a rolling unlabelled session buffer of the test subject's windows), with error guaranteed by Proposition 10.2.

11.5 Phase 1A Diagnostic Metrics, Remediation Ladder, and Fast Smoke-Test Protocol

Because raw loss curves can decrease even when representations degenerate, Phase 1A is monitored at every epoch via five quantitative diagnostics (Table 1).

Remark 11.1 (Ordered Remediation Ladder for Persistent $D_y^{\mathrm{norm}} > 0.5$). Because the Sinkhorn match-to-drop ratio is $\frac{\Pi_{ij}}{d_i} \propto \frac{\exp(-C_{ij}/\varepsilon)}{1-\rho_y}$, if $D_y^{\mathrm{norm}} > 0.5$ persists on a class y after warmup, diagnose and remediate in strict causal order:

1. **Check ε position in $[\varepsilon_{\min}, \varepsilon_{\max}]$ (Theorem 12.2):** If ε sits near $\varepsilon_{\min}$, early cosine costs $C_{ij} \approx 1$ yield $\exp(-C_{ij}/\varepsilon) \ll 1 - \rho_y$, forcing mass into the drop slack d_i. Shift ε toward the midpoint of its admissible interval.

Table 1: Phase 1A Epoch-Level Diagnostic Metrics and Pass/Fail Thresholds.

Diagnostic Metric	**Definition**	**Healthy Target**	**Failure Mode Indicated**
1. Subject Probe Drop	5-fold linear probe predicting subject ID ($1 \dots S$) on z_t vs. h'_t	z_t: 70–90%; h'_t: $<$ 15–20% (near $1/S$ chance)	If $h'_t > 45\%$, rank m is too small or U is stale
2. Zero-Step Val Bag Acc	Nearest-centroid bag accuracy $\arg\min_y \|R_b - \bar{R}_y\|^2$ on held-out subject(s)	Rises from 20% to $\geq 38\%$ once gate g opens	Flat at 20–25% indicates no emotion clustering in $\text{span}(U)^\perp$
3. Nullspace Energy Ratio	$\mathbb{E}[\|h'_t\|^2]/\mathbb{E}[\|z_t - \mu\|^2]$	0.35–0.75	$<$ 0.10 indicates nullspace shrinkage toward τ_{norm}
4. Within-Trial Gate Spread	Intra-bag $\text{Std}_t(\alpha_t)$ and class priors $\rho_1, \dots, \rho_5$	$\text{Std}_t(\alpha_t) > 0.15$; $\rho_{\text{Fear/Disgust}} < \rho_{\text{Neutral}}$	$\text{Std}_t(\alpha_t) < 0.02$ means $\alpha_t \equiv \rho_y$ (uninformative gate)
5. Gate g & OT Drop Rate	Gate g and dropped mass $D_y^{\text{norm}} = \sum_{i \in I_y} d_i$	$g \in [0.4, 0.9]$; $D_y^{\text{norm}} \in [0.1, 0.5]$	$D_y^{\text{norm}} > 0.5$ triggers the 3-step remediation ladder below

2. **Check the ρ_y seed (Section 9.5):** If ε is well-positioned, $D_y^{\text{norm}} > 0.5$ indicates $\tau_y = -\varepsilon \log(1 - \rho_y)$ is too small because ρ_y was under-seeded near $\rho_{\min}$ ($1 - \rho_y \to 1$, making dropping nearly free). Re-verify the Option B fixed-point seed and burst spread constant $s > 1$.

3. **Check Queue Effective Sample Size $\text{ESS}_y = \frac{(\sum_{j \in Q_y} \beta_j)^2}{\sum_{j \in Q_y} \beta_j^2}$ against the prototype richness floor (Section 9.6):** If both ε and ρ_y are calibrated, high dropping means ν_y has collapsed onto a few stale high-β_j outliers ($\text{ESS}_y \ll N_q \tilde{\rho}_{\min}$). Increase N_q toward the ceiling or raise the E-step annealing temperature T.

Remark 11.2 (Fast 5-Subject Smoke-Test Protocol). To verify the complete Phase 1A $\to$ 1B loop in minutes before full 16-subject training, slice by subjects and sessions rather than random windows (fewer than 4 subjects leaves S_{between} rank-deficient):

- **Subset Slice:** Select $S = 5$ training subjects + 1 unseen validation subject, Session 1 only (15 trials/subject = 75 training bags, $\sim$ 3,000 windows, and 15 unseen test bags).
- **Scaled Hyperparameters:** Set subspace rank $m = 3$ ($\leq S - 1 = 4$) and scale queue size proportionally to $N_q = 512$ (batch size $B \in \{64, 128\}$) so the queue turns over every 4–8 steps.
- **Phase 1B Head Protocol:** Evaluate (1) the zero-training nearest-centroid classifier $\hat{y}_b = \arg\min_y \|R_b - \bar{R}_y\|^2$, and (2) a single `nn.Linear(d, 5, bias=True)` layer warm-started at $w_y = \bar{R}_y/\sigma_R^2$, $b_y = -\|\bar{R}_y\|^2/(2\sigma_R^2)$ with L2 weight decay $\lambda_{\text{wd}} \in [10^{-2}, 10^{-1}]$.
- **Smoke-Test Pass Criteria (within 15–20 epochs):** Subject probe drops from $> 80\%$ on z_t to $< 30\%$ on h'_t (20% chance); intra-trial $\text{Std}_t(\alpha_t) > 0.15$; and unseen subject bag accuracy crosses 35–40%+.

12 Analytic Certification (Track B)

This section states the analytic properties of the assembled loss on the convex, compact closed unit ball $\mathbb{B}^d = \{v \in \mathbb{R}^d : \|v\| \leq 1\}$. Full proofs are to be supplied in the Track B appendix using strictly single-variable real analysis (1D derivatives, Mean Value Theorem, Extreme Value Theorem) extended to n terms via finite sums and the triangle inequality.

12.1 Well-Posedness of the Sparsity Term and Baseline Contraction

Proposition 12.1. *The binary divergence* $\mathrm{bkl}(a, \rho) = a\log(a/\rho) + (1-a)\log(\frac{1-a}{1-\rho})$ *satisfies:*

(i) $\mathrm{bkl}(a, \rho) \geq 0$*, with equality iff* $a = \rho$*;*

(ii) continuous on $(0,1)^2$*, continuously extendable in* a *to* $[0,1]$ *for fixed* $\rho \in [\rho_{\min}, 1-\rho_{\min}] \subset (0,1)$*;*

(iii) strictly convex in a *(*$\partial_a^2 \mathrm{bkl}(a,\rho) = \frac{1}{a(1-a)} \geq 4 > 0$*) and strictly increasing in* $|a-\rho|$ *away from* $a = \rho$*;*

(iv) the self-consistency M-step map $\alpha \mapsto \rho_y(\alpha) = \frac{1}{N_y}\sum_{t \in y} \alpha_t$ *is continuous on the compact cube* $[\alpha_{\min}, \alpha_{\max}]^{N_y} \subset (0,1)^{N_y}$*, and both the threshold-free seeding map* $F(u)$ *and the baseline estimator map* $G(\mu')$ *(for* $\kappa_\mu < 1$*) are strict contractions on their compact intervals.*

[Track B Proof Guide (for Nur) — B6 — 1D Calculus Analysis of bkl **and Fixed Points. Step 1 (Non-negativity and convexity in 1D): Fix** $\rho \in (0,1)$ **and treat** $f(a) = a\log(a/\rho) + (1-a)\log(\frac{1-a}{1-\rho})$ **as a single-variable function on** $(0,1)$**. Compute** $f'(a) = \log\frac{a}{1-a} - \log\frac{\rho}{1-\rho}$ **and** $f''(a) = \frac{1}{a} + \frac{1}{1-a} = \frac{1}{a(1-a)} \geq 4 > 0$**. Since** $f''(a) > 0$ **everywhere and** $f'(\rho) = 0$**,** $a = \rho$ **is the unique global minimum with** $f(\rho) = 0$**,** f **is strictly decreasing for** $a < \rho$ **and strictly increasing for** $a > \rho$**.**
Step 2 (Boundary limits): Use standard 1D limit $\lim_{a \to 0^+} a\log a = 0$ **(via L'Hôpital's rule on** $\frac{\log a}{1/a}$**) to show** $f(0^+) = -\log(1-\rho) < \infty$ **and** $f(1^-) = -\log\rho < \infty$**.**
Step 3 (Fixed points): Combine B6(v) and B6(vi) using the 1D Mean Value Theorem $|F(u_1) - F(u_2)| \leq \kappa|u_1 - u_2|$**.]**

12.2 ε-Admissibility Interval

Theorem 12.2. *Let* $\tilde{M} = \max(2, \tau_y)$ *be the maximum entry of the augmented cost matrix* $\tilde{C}$ *on* $\mathbb{B}^d$*. The entropic regulariser* ε *admits the interval*

$$\frac{\tilde{M}}{2\,\mathrm{artanh}(\mathrm{tol}^{1/T})} \leq \varepsilon \leq \varepsilon_{\max},$$

where the lower bound ensures Sinkhorn contraction to tolerance tol *in* T *iterations on the* $(n+1) \times (m+1)$ *augmented kernel* $\tilde{K} = \exp(-\tilde{C}/\varepsilon)$*, and* $\varepsilon_{\max}$ *is determined by the refined blur bound.*

12.3 Definiteness of S_ε

Theorem 12.3. $S_\varepsilon(a,b)$ *is symmetric (*$S_\varepsilon(a,b) = S_\varepsilon(b,a)$*) and satisfies* $S_\varepsilon(a,a) = 0$ *for all* a*. Moreover, in the active matching regime (*$\tilde{C}_{ij} < 2\tau_y$*),* $S_\varepsilon(a,b) \geq 0$ *with* $S_\varepsilon(a,b) = 0 \iff a = b$*.*

12.4 Limit Behaviours

Proposition 12.4. *With the canonical drop price $\tau_y = \varepsilon h(\rho_y) = -\varepsilon \log(1-\rho_y)$:*

(i) $\varepsilon \to 0^+$ *(at fixed $\tau_y > 0$): $W_\varepsilon \to W_0$ (exact unbalanced linear program), uniformly on compact sets.*

(ii) $\varepsilon \to \infty$*: $S_\varepsilon(a,b) \to 0$ at rate $\mathcal{O}(1/\varepsilon)$ as the coupling approaches the product measure.*

(iii) $\rho_y \to 0^+$ *and* $\rho_y \to 1^-$ *(through $h(\rho_y) = -\log(1-\rho_y)$):*

- *As $\rho_y \to 0^+$, $h(0^+) = 0 \implies \tau_y \to 0^+$ (free dropping regime; dummy weight $\exp(-h(0^+)) = 1$).*
- *As $\rho_y \to 1^-$, $h(1^-) = +\infty \implies \tau_y \to +\infty$ (zero dropping regime $d_i = u_j = 0$; dummy weight $\exp(-h(1^-)) = 0$, recovering exact balanced Sinkhorn OT).*

(iv) Gate g: continuous, piecewise linear and Lipschitz with constant $1/\log(K-1)$ away from clamp boundaries, monotone non-decreasing in C_K^{DCL}.

[Track B Proof Guide (for Nur) — B4 — Single-Variable Limit Table.
(i) $\varepsilon \to 0^+$: Since $0 \le -R_\varepsilon(\tilde{\Pi}) \le \varepsilon \log((n+1)(m+1)) + \varepsilon e^{-1}(n+1)(m+1)$, squeeze theorem as $\varepsilon \to 0^+$ gives $\lim_{\varepsilon\to 0^+} W_\varepsilon = W_0$.
(ii) $\varepsilon \to \infty$: Use 1D Taylor expansion $e^{-c/\varepsilon} = 1 - \frac{c}{\varepsilon} + \mathcal{O}(\varepsilon^{-2})$. At $\varepsilon = \infty$, all Gibbs factors equal 1, so $\tilde{\Pi}_{ij}$ is independent of C_{ij} and the self-terms $\frac{1}{2}W_\varepsilon(a,a) + \frac{1}{2}W_\varepsilon(b,b)$ cancel $W_\varepsilon(a,b)$ up to $\mathcal{O}(1/\varepsilon)$.
(iii) $\rho_y \to 0^+, 1^-$: Evaluate 1D limits $\lim_{\rho\to 0^+} -\log(1-\rho) = 0$ and $\lim_{\rho\to 1^-} -\log(1-\rho) = +\infty$.
(iv) Gate g: For $g(x) = \min(1, \max(g_{\mathrm{floor}}, x/\log(K-1)))$, check $|g(x) - g(y)| \le \frac{1}{\log(K-1)}|x-y|$ by checking the 3 intervals $x < g_{\mathrm{floor}} \log(K-1)$, middle, and $x > \log(K-1)$.]

12.5 Non-Degeneracy (Collapse Prevention)

Theorem 12.5 (Non-Degeneracy). *(i) Without $\mathcal{L}_{\mathrm{anc}}$, the constant embedding $\tilde{z}_t \equiv \tilde{z}_0$ is a global minimiser of $\mathcal{L}_{\mathrm{em}} + \mathcal{L}_{\mathrm{sub}}$ (with $\mathcal{L}_{\mathrm{em}} = 0$ and $\mathcal{L}_{\mathrm{sub}} = 0$) and a stationary point of $\mathcal{L}_{\mathrm{em}} + \mathcal{L}_{\mathrm{sub}} + \mathcal{L}_{\mathrm{spr}}$.*

(ii) With $\mathcal{L}_{\mathrm{anc}}$, the constant embedding is not a minimiser: there exists a non-constant configuration in $\mathbb{B}^d$ with strictly smaller $\mathcal{L}_{\mathrm{total}}$.

(iii) For any $c \le c_{\max} < 1$, the anchor gradient strictly dominates the OT gradient along the collapse-escaped direction for all gate values $g \in [g_{\mathrm{floor}}, 1]$.

[Track B Proof Guide (for Nur) — B5 — 1D Perturbation Proof of Non-Degeneracy.
Step 1 (Constant point evaluation): If all points equal $\tilde{z}_0$, every pairwise cost is $C_{ij} = 1 - \|\tilde{z}_0\|^2$ (a constant), so $W_\varepsilon(\mu_y, \nu_y) = W_\varepsilon(\mu_y, \mu_y) = W_\varepsilon(\nu_y, \nu_y)$ and $S_\varepsilon = 0$; similarly $z_t = z_0 \implies \mu_s^\star = \mu = z_0 \implies \mathcal{L}_{\mathrm{sub}} = 0$, and $\alpha_t = \rho_y \implies \mathrm{bkl}(\rho_y, \rho_y) = 0$.
Step 2 (2-point 1D perturbation): Take just two opposite points on a 1D line: $\tilde{z}_1(\delta) = \delta$ and $\tilde{z}_2(\delta) = -\delta$ for $\delta \in [0,1]$, where positive pairs have the same sign and negative pairs have opposite signs. Then the positive score is $+\delta^2/\tau_{\mathrm{DCL}}$ and negative score is $-\delta^2/\tau_{\mathrm{DCL}}$. Differentiating this 1D function of $v = \delta^2$ at $v = 0$ gives a strictly negative slope for $\mathcal{L}_{\mathrm{DCL}}$, while the $c_{\max} < 1$ bound from B5(iii) ensures $\mathcal{L}_{\mathrm{total}}(\delta) < \mathcal{L}_{\mathrm{total}}(0)$ for small $\delta > 0$.]

12.6 Boundedness and Lipschitz Continuity of All Terms on $\mathbb{B}^d$

Theorem 12.6. *On the convex, compact closed unit ball domain $\tilde{z}_t \in \mathbb{B}^d$ ($\|\tilde{z}_t\| \le 1$, $\|z_t\| \le 1$) and interior responsibilities $\alpha_t \in [\alpha_{\min}, \alpha_{\max}] \subset (0,1)$, each of $\mathcal{L}_{\text{em}}$, $\mathcal{L}_{\text{sub}}$, $\mathcal{L}_{\text{spr}}$, the gate g, and the length-normalised readout map $R_b = \frac{1}{T_b}\sum_{t=1}^{T_b} \alpha_t^{(1\text{B})} h'_t$ is bounded and Lipschitz continuous with explicit T_b-independent constants:*

- $\|\nabla_{\tilde{z}_i} W_\varepsilon^y\| \le 1$ *and* $\|\nabla_{\tilde{z}_i} S_\varepsilon^y\| \le 2$ *(and with respect to h'_i, bounded by $1/\tau_{\text{norm}} = 4$ and $2/\tau_{\text{norm}} = 8$);*
- $\|\nabla_{z_t} \mathcal{L}_{\text{sub}}\| \le 4$;
- $|\partial_{\alpha_t} \text{bkl}(\alpha_t, \rho_y)| \le \log \frac{\alpha_{\max}(1-\rho_{\min})}{\alpha_{\min}\rho_{\min}}$;
- $\|R_b\| \le 2$ *and* $\text{Lip}(R_b) \le 1$ *with respect to the average window perturbation* $\frac{1}{T_b}\sum_{t=1}^{T_b} \|\Delta h'_t\|$.

[Track B Proof Guide (for Nur) — B7 — 1D Line-Segment Proof of Boundedness and Lipschitz Constants on $\mathbb{B}^d$.
Why $\mathbb{B}^d$ makes this pure 1D calculus: Because the closed unit ball $\mathbb{B}^d = \{v : \|v\| \le 1\}$ is convex, for any two points $x, y \in \mathbb{B}^d$ the straight line segment $\gamma(s) = (1-s)x + sy$ stays inside $\mathbb{B}^d$ for all $s \in [0,1]$!
Step 1 (1D Mean Value Theorem): For any differentiable function $\mathcal{L}$ on $\mathbb{B}^d$, define the single-variable function $\phi(s) = \mathcal{L}((1-s)x + sy)$ for $s \in [0,1]$. By the 1D Mean Value Theorem, $|\mathcal{L}(y) - \mathcal{L}(x)| = |\phi(1) - \phi(0)| = |\phi'(s^\star)| \le \max_{s\in[0,1]} \|\nabla\mathcal{L}(\gamma(s))\| \cdot \|y - x\|$.
Step 2 (Plug in the 4 derivative bounds):
1. For $\mathcal{L}_{\text{em}}$: Proposition 8.1 gives $\|\nabla_{\tilde{z}_i} W_\varepsilon^y\| \le 1$ and $\|\nabla_{\tilde{z}_i} S_\varepsilon^y\| \le 2$ on $\mathbb{B}^d$.
2. For $\mathcal{L}_{\text{sub}}$: in 1D, $\frac{d}{dz}(1-\alpha)(z-\mu_s)^2 = 2(1-\alpha)(z-\mu_s)$, and since $|1-\alpha| \le 1$ and $|z - \mu_s| \le |z| + |\mu_s| \le 2$, the derivative is at most $2 \times 1 \times 2 = 4$.
3. For $\mathcal{L}_{\text{spr}}$: from B6, $|f'(a)| = \left|\log \frac{a(1-\rho)}{(1-a)\rho}\right| \le \log \frac{\alpha_{\max}(1-\rho_{\min})}{\alpha_{\min}\rho_{\min}}$ on $[\alpha_{\min}, \alpha_{\max}]$.
4. For R_b: by the triangle inequality on a finite sum, $\left\|\frac{1}{T_b}\sum_{t=1}^{T_b} \alpha_t(h'_{t,1} - h'_{t,2})\right\| \le \frac{1}{T_b}\sum_{t=1}^{T_b} |\alpha_t| \, \|h'_{t,1} - h'_{t,2}\| \le \frac{1}{T_b}\sum_{t=1}^{T_b} \|h'_{t,1} - h'_{t,2}\|$.]

13 Failure Catalogue

14 Conclusion

We have shown that the assembled Phase-1A/1B objective of SWEEP-Net is the ELBO of a latent-variable model in which emotion prototypes carry no subject index and subject baselines carry no emotion index. Optimising the loss in a single unified embedding space with a pure projected anchor on $\mathbb{B}^d$ is, by construction, a solution to subject-ID memorisation and temporal label pollution. Phase 1B's length-normalised readout $R_b = \frac{1}{T_b}\sum_{t=1}^{T_b} \alpha_t^{(1\text{B})} h'_t$, evaluated via the closed-form label-free nullspace marginal E-step (14) (with a certified two-step contraction baseline estimator) and classified by a single linear Bayes plug-in head $\ell_y(R_b) = w_y^\top R_b + b_y$, is the deterministic sufficient statistic of the same model—direction is emotion content, magnitude is burst rate, and subject signal is projected out for both seen and unseen subjects.

The loss is certified as well-defined, bounded, continuous, non-degenerate, and equipped with a provably admissible range for ε and canonical drop price $\tau_y = -\varepsilon \log(1 - \rho_y)$ by the accompanying real-analysis appendix (Track B).

Dropped or Altered Component	Pathology That Returns
Remove $\mathcal{L}_{\text{anc}}(\tilde{z})$ (Projected DCL)	Constant-vector or nullspace collapse (cMMD dead config)
Remove unbalanced dropping (d_i, u_j)	Balanced OT $\to$ SCDA subject mining (87% probe)
Remove nullspace projection $(\mathcal{L}_{\text{sub}}, I - UU^\top)$	Subject memorisation returns; probe $\to$ 91%
Split 1A queue (128-D) vs. 1B readout (256-D)	Backbone stashes subject ID in MLP head nullspace; gate/readout mismatch
Evaluate $\log p_s$ on full space instead of h'_t	Within-trial jitter in span(U) falsely triggers burst gate α_t
Remove sparsity $(\mathcal{L}_{\text{spr}})$	ρ_y unconstrained; gaslighting on Fear/Disgust
Replace exact E-step with learned α-head	Calibration failure; OT activates on uncalibrated gate
Omit $1/T_b$ normalisation or bias b_y in 1B head	Trial-duration bias (T_b) or loss of burst-rate magnitude (ρ_y)
Back-prop through Sinkhorn fixed point	NaN blowup (DANN/GRL dead config)
Queue-vs-queue OT	Zero gradient; dead term

The paper sentence. *"What we were optimising all along is the ELBO of a subject-conditional, burst-gated latent model; the assembled contrastive, transport, projection, and sparsity terms are its four faces, and Phase 1B reads out the sufficient statistic."*

Appendix: Track B Proof Obligations (1D-First Guide for Nur)

Every proof obligation below is designed so that **no linear algebra or measure theory is needed**: you always prove the statement first for **1D numbers (or a 2-point pair)** using single-variable calculus (derivatives, $\log t \le t - 1$, Mean Value Theorem, Extreme Value Theorem on $[a, b]$), and then extend it to n terms by writing a finite sum $\sum_{i=1}^{n}$ and applying the basic triangle inequality $|x + y| \le |x| + |y|$.

B1. Weighted Jensen bound (Theorem 4.1): Prove $\log t \le t - 1$ for $t > 0$ using $g'(t) = \frac{1}{t} - 1$, substitute $t = \frac{A_e/S}{q_e}$, multiply by q_e, and sum over e.

B2. ε-admissibility (Theorem 12.2): (B2a) Differentiate the 1D log-ratio map $f(u) = \log \frac{e^{u-\Delta}+1}{e^{u}+e^{-\Delta}}$ to show $0 < f'(u) \le \tanh(\Delta) < 1$ where $\Delta = \tilde{M}/(2\varepsilon)$, and solve $[\tanh(\tilde{M}/2\varepsilon)]^T \le \text{tol}$; (B2b) apply B1 with $N = (n+1)(m+1)$ uniform weights; (B2c) bound a 2-bin mass shift $|a_1 - b_1|$ and sum over bins.

B3. Definiteness of S_ε (Theorem 12.3): (i)–(ii) Check $S_\varepsilon(a, a) = 0$ and $S_\varepsilon(a, b) = S_\varepsilon(b, a)$ from Definition 5.4; (iii) analyse the 1D function $\psi_c(p)$ for matching two single points at cost $c \in [0, 2]$ vs. dropping at cost $2\tau_y$, showing $S_0 = \min(c, 2\tau_y) > 0$ for $0 < c < 2\tau_y$ and flat saturation at $2\tau_y$ for $c \ge 2\tau_y$.

B4. Limit behaviours (Proposition 12.4): Compute the 1D limits $\varepsilon \to 0^+$ (squeeze theorem), $\varepsilon \to \infty$ (Taylor expansion $e^{-c/\varepsilon} \approx 1 - c/\varepsilon$), $\rho_y \to 0^+$ $(h(0^+) = 0)$, $\rho_y \to 1^-$ $(h(1^-) = +\infty)$, and the 3-piece slope of clamp.

B5. Non-degeneracy and $c_{\max} < 1$ (Theorem 12.5): Evaluate the loss at constant $\tilde{z}_0$, then take the 1D derivative $\phi'(0) \le -(1 - cg)A < 0$ along a 1D line away from collapse to show $c \le c_{\max} < 1$.

B6. Binary divergence bkl **and Fixed-Point Contractions (Propositions 12.1 & 10.2):** Differentiate $f(a) = \mathrm{bkl}(a, \rho)$ twice in 1D ($f''(a) = \frac{1}{a(1-a)} \geq 4 > 0$), and apply the 1D quotient rule + Mean Value Theorem to certify both $F(u)$ (for ρ_y) and the two-step baseline map $G(\mu)$ (for $\hat{\mu}'_b$).

B7. Boundedness and Lipschitz constants on $\mathbb{B}^d$ (Theorem 12.6): Connect any two points $x, y \in \mathbb{B}^d$ by the 1D line segment $\gamma(s) = (1-s)x + sy$ ($s \in [0,1]$) and apply the 1D Mean Value Theorem $|\phi(1) - \phi(0)| \leq \max |\phi'(s)|$ to the four explicit derivative bounds.

References

[1] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *JRSS-B*, 39(1):1–38, 1977.

[2] C. M. Bishop. *Pattern Recognition and Machine Learning*, Ch. 10. Springer, 2006.

[3] K. Rose. Deterministic annealing for clustering, compression, classification, regression, and related optimization problems. *Proc. IEEE*, 86(11):2210–2239, 1998.

[4] M. Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. *NeurIPS*, 2013.

[5] L. Chizat, G. Peyré, B. Schmitzer, and F.-X. Vialard. An interpolating distance between optimal transport and Fisher–Rao metrics. *Found. Comput. Math.*, 18:1–44, 2018.

[6] A. van den Oord, Y. Li, and O. Vinyals. Representation learning with contrastive predictive coding (InfoNCE). *arXiv:1807.03748*, 2018.

[7] C.-H. Yeh et al. Decoupled contrastive learning. *ECCV*, 2022.